AI-Scientist: A Multi-Agent Orchestrated Framework for Scientific Claim Verification and Adaptive Reasoning

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AI-Scientist: A Multi-Agent Orchestrated Framework for Scientific Claim Verification and Adaptive Reasoning

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Dedicated to everyone whose patience, support, and belief made this work possible.

Approval

This thesis titled **AI-Scientist: A Multi-Agent Orchestrated Framework for Scientific Claim Verification and Adaptive Reasoning** has been reviewed in draft form and is approved as a candidate dissertation for the degree of **Master of Technology** in **Computer Science and Engineering**.

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Declaration

I hereby declare that the work reported in this thesis, entitled **AI-Scientist: A Multi-Agent Orchestrated Framework for Scientific Claim Verification and Adaptive Reasoning** , is an original record of research carried out by me under the supervision of **Mr. K.G. Atram**. To the best of my knowledge, this work has not been submitted, either in part or in full, to any other institution for the award of any degree or diploma.

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Abstract

Large language models can answer scientific questions fluently, but they frequently assert claims that are not grounded in evidence, fabricate or misattribute citations, and offer no transparent account of how a conclusion was reached. This thesis presents **AI-Scientist**, a multi-agent orchestrated framework for scientific claim verification and adaptive reasoning that addresses these weaknesses by separating the work of a research assistant into specialized, cooperating agents rather than entrusting it to a single monolithic model.

In AI-Scientist, an Orchestrator coordinates a Retrieval agent, a Claim Extraction agent, a Verification agent, a Critic agent, a Synthesis agent, and a Report agent. Given a research question from any academic domain, the system grounds its reasoning in evidence drawn from three layers — user-uploaded papers, a curated offline corpus, and live scholarly sources retrieved from PubMed Central and arXiv — and produces an evidence-backed verdict for each extracted claim, an explicit confidence estimate, and a traceable record of the reasoning steps that led to the answer. A central design commitment is that the system should report *insufficient evidence* rather than overclaim, and should surface contradictions instead of hiding them.

The framework is studied against two baselines that represent the conventional alternatives: a direct single-agent reader and a standard retrieval-augmented generation pipeline. The thesis argues, and the system demonstrates, that an orchestrated multi-agent design improves verification quality and reasoning traceability along several axes: it grounds conclusions in retrieved evidence, it makes its uncertainty explicit through calibrated confidence and an iterative critic-driven re-verification loop, and it favours dependable, source-attributed answers over confident but unsupported ones. AI-Scientist is therefore positioned not as a system that replaces human scientific judgement, but as a cautious, evidence-aware collaborator that makes the basis for every conclusion inspectable.

Keywords: multi-agent systems, scientific claim verification, retrieval-augmented generation, hallucination mitigation, evidence synthesis, large language models.

Contents

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
arXiv	Open-access preprint repository for physics, mathematics, and computer science
CS	Computer Science
JATS	Journal Article Tag Suite (PubMed Central XML format)
JSON	JavaScript Object Notation
LLM	Large Language Model
NCBI	National Center for Biotechnology Information
PMC	PubMed Central
RAG	Retrieval-Augmented Generation
RQ	Research Question
UI	User Interface

List of Symbols and Notations

q	Research question or claim submitted by the user.
d	A candidate document (paper title, abstract, and topics).
$T(\cdot)$	Set of normalized content tokens of a text.
$\mathrm{cov}(q, d)$	Query coverage: fraction of query tokens present in document d.
$\mathrm{ovl}(a, b)$	Jaccard overlap between the token sets of texts a and b.
b_s	Source ranking bonus for source s (uploaded, corpus, or live API).
τ_{cov}	Minimum query-coverage threshold for retrieval.
c	Confidence score assigned to a claim verdict, with $c \in [c_{\min}, c_{\max}]$.
$c_{\min}$, $c_{\max}$	Confidence floor and ceiling enforced by the verifier.
K	Number of papers retrieved per pass (top-K).

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CHAPTER 1
Introduction

Preface

This chapter introduces the AI-Scientist thesis, motivates the problem of unverified and hallucinated scientific claims in large language models, defines the core research gap, and states the thesis contribution around multi-agent orchestration, evidence grounding, and honest, traceable verification.

Large language models (LLMs) have rapidly become a default interface for asking questions about the scientific literature. A researcher, student, or clinician can now type a natural-language question and receive a fluent, confident answer in seconds. This fluency is also the source of a serious problem. The same models that summarize a field convincingly will, with equal confidence, assert findings that no study supports, attribute results to papers that do not contain them, and invent citations that look entirely plausible. In settings where the cost of a wrong answer is high — medicine, biology, public policy, or any domain where decisions follow from evidence — a confident but ungrounded answer can be worse than no answer at all.

The difficulty is not only that models occasionally make mistakes. It is that a single-model answer offers the reader no way to tell a grounded conclusion from a fabricated one. The output arrives as a finished paragraph with no visible separation between what was retrieved, what was inferred, and what was simply generated. There is no explicit confidence, no list of the evidence that was actually consulted, and no record of the reasoning steps that produced the conclusion. For a tool that is meant to support research, this opacity is the central failure mode. The question this thesis addresses is therefore not "can an LLM answer a scientific question?" but rather "how can a system answer a scientific question in a way that is grounded, calibrated, and inspectable?"

This thesis presents **AI-Scientist**, a multi-agent orchestrated framework for scientific claim verification and adaptive reasoning. Instead of relying on one model to read, reason, and conclude in a single opaque step, AI-Scientist decomposes the task into specialized agents coordinated by an Orchestrator: a Retrieval agent gathers relevant papers, a Claim Extraction agent turns them into discrete checkable statements, a Verification agent weighs each statement against evidence, a Critic agent challenges weak or single-source conclusions, a Synthesis agent composes a grounded summary, and a Report agent assembles the final evidence-backed output with confidence scores and a reasoning trace. The design treats verification, rather than generation, as the primary task.

1.1 Research Context and Gap

The research landscape already contains strong prior work on retrieval-augmented generation (RAG), on automated fact-checking, and on multi-agent LLM systems. Retrieval-augmented approaches reduce hallucination by conditioning generation on retrieved passages, and a large body of work studies how to fetch, rank, and integrate that context. Multi-agent and "orchestrator" designs have shown that decomposing a task across cooperating models can improve robustness on complex reasoning problems. AI-Scientist does not claim to be the first system to retrieve evidence before answering, nor the first to use multiple cooperating agents. Such claims would be inaccurate and would obscure the actual research opportunity.

The genuine gap addressed by this thesis is different. Most retrieval-augmented assistants still optimize for a single fluent answer and treat retrieval as a silent preprocessing step; they rarely expose, per claim, whether the evidence actually supports the statement, how strongly, from how many independent sources, and what should be done when the evidence is thin or contradictory. In other words, the literature has established that grounding helps, but it has not adequately framed scientific question answering as an explicit, auditable *claim-by-claim verification process* with calibrated confidence and a first-class option to abstain. This is where AI-Scientist contributes: it adds an orchestrated verification and critique layer on top of an already-established grounding setting, and it makes that layer the visible product rather than an implementation detail.

1.2 Thesis Position

AI-Scientist demonstrably answers research questions end to end: given a question from any academic domain, it retrieves evidence, extracts claims, assigns a grounded verdict and confidence to each, and produces a report in which every conclusion is traceable to its sources. The position taken in this thesis builds on that capability rather than overstating it. AI-Scientist is claimed as a reproducible, evidence-grounded, verification-first research assistant that is dependable when the evidence is clear and appropriately cautious when it is not. The main contribution is consequently not raw answer fluency or breadth of coverage; it is the engineering of *trustworthy* scientific answering — knowing *when to say "insufficient evidence"* — together with an orchestration loop in which a critic can force targeted re-verification before a conclusion is treated as settled.

The framing of the project rests on four commitments that recur throughout the thesis:

1. **Grounding over generation.** Every conclusion must be tied to specific retrieved evidence. A claim with no supporting sentence in the evidence pool is reported as unsupported, not paraphrased into apparent fact.

2. **Calibrated confidence over false certainty.** The system attaches an explicit

confidence to each verdict, weighted by evidence strength, corroboration, and source independence, and it deliberately reserves the highest confidence for genuinely well-supported claims.

3. **Critique over single-pass acceptance.** A dedicated Critic agent flags insufficient, low-confidence, single-source, and contradicted claims, and can trigger a second retrieval-and-verification pass aimed precisely at the weak claim.

4. **Traceability over opacity.** The pipeline records an orchestration trace and per-claim evidence so that a reader can reconstruct exactly why the system reached each conclusion.

These commitments are valuable precisely because they constrain the system. A central strength of AI-Scientist is that it is designed to under-claim rather than over-claim: when evidence is missing or conflicting, it says so.

1.3 Problem Motivation

The motivation for AI-Scientist comes from the mismatch between how scientific questions are actually settled and how single-model assistants answer them. In real research practice, a claim is not accepted because it sounds authoritative; it is accepted because it is supported by evidence, ideally corroborated across independent sources, and it remains open to challenge when contradictory findings appear. A useful research assistant therefore needs at least three properties.

First, it must be **evidence-grounded**. Scientific answers cannot be trusted on the strength of phrasing alone. Retrieved papers, their abstracts, and the specific sentences that bear on the question provide the material against which a claim must be checked.

Second, it must be **uncertainty-aware**. A generated statement should not be treated as established merely because it is plausible. Practical verification demands an explicit notion of confidence, an account of how many independent sources agree, and a willingness to report uncertainty honestly.

Third, it must be **self-critical**. The system must be able to recognize when a conclusion rests on a single source, when confidence is low, or when the evidence is contradictory, and it must be able to act on that recognition by seeking more evidence rather than papering over the gap.

AI-Scientist is designed around these three requirements. The resulting system is not a general-purpose oracle; it is a scoped verification assistant whose primary value lies in disciplined reasoning under noisy and incomplete evidence.

1.4 Research Questions

This thesis is organized around the following research questions:

1. Can a multi-agent orchestrated framework improve scientific claim verification and reasoning traceability compared with a direct single-agent reader and a standard RAG pipeline?
2. Does grounding answers in a layered evidence base — user-uploaded papers, a curated corpus, and live scholarly sources — reduce unsupported claims relative to ungrounded answering?
3. Can per-claim confidence, weighted by evidence strength and source independence, give the user a usable signal for when to trust a conclusion?
4. Does an explicit critic-driven re-verification loop improve the handling of weak, single-source, or contradicted claims?
5. Can the system remain domain-general, accepting research questions across fields while still rejecting clearly non-research queries?

These questions keep the work grounded. They separate retrieval from verification, verification from critique, and confidence from certainty. They also ensure that the thesis does not collapse into a vague "AI for science" claim.

1.5 Contributions

The contributions of this thesis are summarized as follows:

1. **A system contribution:** AI-Scientist, an end-to-end multi-agent framework that integrates retrieval, claim extraction, verification, critique, synthesis, and reporting behind a single Orchestrator, with a Streamlit interface and a programmatic API.
2. **A methodological contribution:** a verification-first formulation of scientific question answering, in which the unit of output is a checked claim with a verdict, an explicit confidence, and attributed evidence, rather than a single undifferentiated paragraph.
3. **An evidence-layering contribution:** a layered, source-weighted retrieval strategy that combines user-uploaded papers, a curated offline corpus, and live results from PubMed Central and arXiv, with an explicit priority that favours user-provided and freshly retrieved evidence.
4. **An orchestration contribution:** an iterative critic-driven re-verification loop that targets weak claims specifically, together with a transparent orchestration trace that records how each conclusion was reached.
5. **A comparative framing:** an explicit positioning against single-agent and RAG baselines that clarifies what the multi-agent design adds and where its limits lie.

1.6 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews the background and related work on LLM hallucination, retrieval-augmented generation, automated fact-checking, and multi-agent reasoning systems. Chapter 3 defines the problem, the scope, and the requirements of AI-Scientist as a verification system. Chapter 4 presents the system design and architecture, including the orchestrated agent pipeline and the layered evidence model. Chapter 5 describes the implementation of the agents and their supporting modules. Chapter 6 explains the evaluation methodology, baselines, and the qualities the system is judged on. Chapter 7 reports the evaluation through representative cross-domain case studies and a comparison with the baselines. Chapter 8 discusses the broader meaning of these findings. Chapter 9 examines threats to validity and current limitations. Chapter 10 concludes the thesis and outlines future research directions.

1.7 Chapter Conclusion

This chapter has introduced the AI-Scientist thesis, its motivation, research questions, and contributions. It has also established the central framing used throughout the document: answering a scientific question well is not a generation problem but a verification problem that requires evidence grounding, calibrated confidence, explicit critique, and traceable reasoning. The next chapter situates this framing within the related literature.

CHAPTER 2

Background and Related Work

Preface

This chapter reviews the research context for AI-Scientist, including LLM hallucination and its measurement, retrieval-augmented generation, automated scientific fact-checking, and multi-agent reasoning and self-critique systems.

2.1 Introduction

The AI-Scientist problem sits at the intersection of several active research areas rather than belonging to only one of them. It is related to the study of hallucination in large language models, to retrieval-augmented generation, to automated fact-checking and scientific claim verification, and to the growing literature on multi-agent reasoning and self-critique [1, 2, 3, 4]. A good related-work chapter for AI-Scientist therefore needs to do more than list nearby papers. It must explain which parts of the prior literature AI-Scientist inherits directly, which capabilities have already been established by others, and which gap still remains meaningful enough to support a thesis contribution.

As of the time of writing, the field has moved well beyond the point where one can claim novelty simply by saying that a model retrieves evidence before answering. Retrieval-augmented and self-reflective systems already demonstrate that grounding and self-critique reduce hallucination. The important question is no longer whether grounding should be used at all, but rather how scientific answering should be *structured*, *verified*, and *made transparent*. This chapter reviews the most relevant prior work and shows why AI-Scientist is best positioned not as "the first grounded research assistant," but as an orchestrated, verification-first, evidence-attributed system built on top of an already established grounding setting.

2.2 Hallucination in Large Language Models

The motivating problem for AI-Scientist is hallucination: the tendency of generative models to produce statements that are fluent and confident but unsupported by, or contradictory to, the underlying facts. Surveys of the phenomenon distinguish between intrinsic hallucination, where the output contradicts the provided source, and extrinsic hallucination, where the output cannot be verified against any available source [1, 5]. Both are damaging in a scientific setting, but extrinsic hallucination is especially dangerous because it includes fabricated findings and invented citations that a non-expert reader cannot easily detect.

This literature matters for AI-Scientist in two ways. First, it establishes that hallucination is not a rare edge case but a structural property of how current models generate text, which justifies building verification into the system rather than trusting the base model. Second, it motivates the central design decision of the thesis: to make the evidence for each statement explicit, so that intrinsic hallucination becomes checkable against the retrieved sources and extrinsic hallucination is surfaced as "insufficient evidence" rather than presented as fact.

A related strand of work focuses on *detecting* hallucination after generation. SelfCheckGPT, for example, samples multiple responses and measures their consistency as a black-box signal of factuality, on the intuition that hallucinated content varies more across samples than grounded content [6]. AI-Scientist shares the underlying goal of flagging untrustworthy statements, but takes a complementary route: rather than detecting hallucination by self-consistency alone, it grounds each claim in retrieved evidence and assigns confidence from the strength and independence of that evidence.

2.3 Measuring Factual Precision

Detecting hallucination is only useful if factuality can be measured at a meaningful granularity. Coarse, response-level judgments hide the fact that a long answer may be partly correct and partly fabricated. FActScore addresses this by decomposing a generated passage into atomic facts and scoring each against a knowledge source, yielding a fine-grained precision measure [7]. This decomposition principle is directly relevant to AI-Scientist, whose Claim Extraction agent performs an analogous step: it breaks the retrieved material into discrete, individually checkable claims so that verification and confidence can be reported per claim rather than per answer.

The lesson AI-Scientist takes from this line of work is methodological. A trustworthy research assistant should not output a single monolithic verdict on a whole answer. It should expose the unit of evidence at the level of individual claims, because that is the level at which support, contradiction, and uncertainty actually differ.

2.4 Retrieval-Augmented Generation

The closest architectural ancestor of AI-Scientist is retrieval-augmented generation. The original RAG formulation conditions a generator on passages retrieved from an external corpus, improving performance on knowledge-intensive tasks and reducing reliance on parametric memory [8]. Subsequent surveys document a rich design space covering how to index, retrieve, rerank, and fuse evidence, and how to balance retrieval cost against answer quality [2]. AI-Scientist inherits the core RAG insight that grounding generation in retrieved text reduces fabrication, and its Retrieval agent and layered evidence model are squarely within this tradition.

However, standard RAG also sets a baseline that AI-Scientist must improve upon rather than merely reproduce. In a typical RAG pipeline, retrieval is a silent preprocessing step and the model still emits a single fused answer. The reader sees neither which passages actually supported each part of the answer nor how confident the system is in each component. Self-RAG sharpens this picture by training a model to decide when to retrieve and to critique its own generations with reflection tokens, moving RAG from a fixed pipeline toward an adaptive, self-evaluating one [9]. AI-Scientist pursues the same adaptive spirit, but realizes it through an explicit multi-agent division of labour and an external critic rather than through learned reflection tokens, which keeps the verification logic inspectable and model-agnostic.

For AI-Scientist, the most useful interpretation of the RAG literature is that grounding is necessary but not sufficient. Retrieval supplies the raw material for verification; it does not, on its own, tell the user which claims are well-supported, which rest on a single source, and which are contradicted.

2.5 Automated Fact-Checking and Scientific Claim Verification

A second direct ancestor is automated fact-checking, and in particular scientific claim verification. The FEVER task framed fact verification as retrieving evidence sentences and labeling a claim as supported, refuted, or lacking information [10]. SciFact specialized this setting to scientific claims, requiring a system to find abstracts that support or contradict a claim and to identify the specific rationale sentences [3]. This labeling scheme — supported, contradicted, or insufficient evidence — maps almost directly onto the verdict space used by the AI-Scientist Verification agent.

This correspondence is important for positioning. AI-Scientist does not claim to invent the idea of labeling a scientific claim against retrieved evidence; that task is well established. What the fact-checking literature generally does not provide is an end-to-end *assistant* that takes a free-form research question, extracts its own claims from retrieved papers, verifies them, criticizes the weak ones, and synthesizes a confidence-annotated report for a human user across arbitrary domains. AI-Scientist is best understood as embedding the claim-level verification paradigm inside a usable, orchestrated research workflow.

The fact-checking tradition also reinforces a key design constraint: the evidence rationale must be explicit. A verdict without the supporting sentence is not auditable. AI-Scientist therefore attaches, to each verdict, the specific evidence snippets and their overlap with the claim, mirroring the rationale-selection requirement of scientific claim verification.

2.6 Reasoning, Self-Critique, and Verification Loops

A third strand of related work concerns how language-model systems reason and check themselves over multiple steps. ReAct interleaves reasoning traces with actions such as retrieval, improving grounding and interpretability [11]. Reflexion adds a verbal self-feedback loop in which an agent reflects on a failed attempt and retries with that feedback [12]. Chain-of-Verification has a model draft an answer, generate verification questions, answer them independently, and revise, explicitly reducing hallucination through a self-checking loop [13].

These systems are conceptually close to the AI-Scientist Critic agent and its re-verification loop. The shared idea is that a first attempt should not be trusted as final, and that a structured second look can catch errors the first pass missed. AI-Scientist differs in where it places the critique. Rather than asking one model to critique itself — which risks the critique inheriting the same blind spots as the original answer — it assigns critique to a separate agent with explicit rules for flagging insufficient, low-confidence, single-source, and contradicted claims, and it routes the resulting re-verification at the granularity of the individual weak claim. This makes the loop transparent and targeted rather than global and opaque.

2.7 Multi-Agent Frameworks

The final and most architecturally relevant strand is the recent wave of multi-agent LLM frameworks. AutoGen provides infrastructure for multiple conversational agents that cooperate and use tools [4]; CAMEL studies role-playing communicative agents [14]; and multi-agent debate shows that having several model instances argue and converge can improve factuality and reasoning over a single instance [15]. Orchestration libraries such as LangGraph make it practical to express such systems as explicit state machines over cooperating actors [16].

This environment creates both a risk and an opportunity for AI-Scientist. The risk is that a vague claim such as "we built a multi-agent system" will sound derivative, because multi-agent scaffolding is now common. The opportunity is that most multi-agent demonstrations emphasize open-ended collaboration or debate, whereas AI-Scientist uses a small, fixed set of specialized roles wired into a deterministic orchestration with a clear purpose: verify claims against layered evidence and report the result honestly. The contribution is therefore not the existence of multiple agents, but the disciplined, verification-oriented way they are composed.

The debate literature is particularly instructive. Multi-agent debate improves factuality by forcing disagreement to surface before consensus [15]. AI-Scientist obtains a related benefit more cheaply: instead of several generalist agents debating, a single Critic agent applies explicit weakness criteria and can demand more evidence, which provides much of the value of adversarial review without the cost and unpredictability of free debate.

2.8 Evaluating Grounded Systems

Because AI-Scientist makes claims about grounding and trustworthiness, it must be read alongside the literature on how such systems are evaluated. Frameworks such as RAGAS propose reference-light metrics for retrieval-augmented systems, including faithfulness (whether the answer is entailed by the retrieved context), answer relevance, and context relevance [17]. These dimensions correspond closely to the qualities AI-Scientist is designed to optimize: faithfulness maps to evidence grounding, and context relevance maps to the quality of retrieval ranking.

The evaluation literature also cautions that grounded systems are easy to over-credit. A system can appear faithful simply by hedging, or appear confident simply by ignoring contradictory evidence. This caution shapes the evaluation philosophy adopted later in this thesis, which emphasizes whether the system grounds, calibrates, and exposes its reasoning, rather than only whether it produces an agreeable-sounding answer.

2.9 Infrastructure: Scholarly Retrieval Sources

Finally, AI-Scientist depends on concrete scholarly retrieval infrastructure. Live evidence is drawn from PubMed Central through the NCBI E-utilities programming interface [18] and from arXiv through its public API [19]. Classical lexical relevance ideas, exemplified by the BM25 family of ranking functions [20], inform the token-overlap and coverage measures used to rank candidate evidence. These are not contributions of the thesis, but they are essential building blocks, and the way AI-Scientist combines them with user uploads and a curated corpus is part of its evidence-layering design.

2.10 Implications for AI-Scientist Design

Taken together, the literature suggests several design consequences for AI-Scientist.

First, grounding is necessary but insufficient. RAG reduces hallucination, but the user still needs per-claim support, confidence, and contradiction information that a fused answer does not provide.

Second, factuality should be measured at the claim level. FActScore and scientific claim verification both argue for decomposing answers into atomic, individually checkable units.

Third, verdicts must carry their evidence. The fact-checking tradition shows that a label without a rationale is not auditable.

Fourth, self-critique helps, but it is most trustworthy when it is structured and external rather than left to the same model that produced the answer.

Fifth, multi-agent orchestration is valuable not for its novelty but for the discipline it imposes: a fixed set of specialized roles with a clear verification purpose is easier to reproduce, inspect, and reason about than free-form agent collaboration.

2.11 Chapter Conclusion and Positioning

The literature reviewed in this chapter leads to a clear and disciplined positioning for AI-Scientist. Prior work has already demonstrated that grounding reduces hallucination, that factuality can be measured at the claim level, that scientific claims can be labeled against retrieved abstracts, and that self-critique and multi-agent collaboration can improve reasoning. At the same time, prior work also makes clear that standard grounded systems still tend to emit single fused answers without exposing per-claim support, confidence, and disagreement to the user.

AI-Scientist therefore does not claim novelty in grounding or in the use of multiple agents. Its contribution lies in composing these established ideas into a *verification-first, evidence-attributed, self-critical research assistant* in which every conclusion is a checked claim with a calibrated confidence and a traceable rationale. That framing is supported, not contradicted, by the literature. The next chapter turns from this background to the formal problem definition, requirements, and scope of the AI-Scientist system.

CHAPTER 3

Problem Statement, Requirements, and Scope

Preface

This chapter formalizes the AI-Scientist task, defines the verdict space and the layered evidence model, states the verification and honesty requirements, and fixes the practical scope of the thesis so later architecture and evaluation claims remain precise and defensible.

3.1 Introduction

The previous chapter established that grounded scientific answering is already a recognized research task, but that the most defensible remaining gap lies in *verification-first*, *evidence-attributed*, and *honest* answering. This chapter turns that positioning into a precise problem statement. Rather than treating AI-Scientist as a generic question-answering agent, the thesis defines it as a constrained verification system: it observes a research question, gathers layered evidence, decomposes the relevant material into discrete claims, assigns each claim a grounded verdict and confidence, criticizes the weak claims, and reports the result with a traceable rationale.

This precision matters for two reasons. First, it prevents overclaiming. A thesis that vaguely states "AI-Scientist answers research questions" would blur together retrieval, claim extraction, verification, and synthesis. Second, a well-defined problem statement makes the evaluation meaningful. The qualities examined later in the thesis measure not only whether an answer is produced, but whether the answer is grounded, calibrated, and inspectable.

3.2 Task Definition

AI-Scientist addresses the following task: given a natural-language research question q from any academic domain, and an evidence environment consisting of optional user-uploaded papers, a curated offline corpus, and optional live scholarly sources, produce a structured report that contains a grounded conclusion supported by traceable evidence. The task is therefore not merely "generate an answer." It is a structured verification problem over noisy and incomplete evidence.

At a high level, the system input consists of:

1. the research question or claim q submitted by the user,

2. an optional set of user-uploaded papers supplied for the session, and
3. an evidence environment: a curated corpus that is always available offline, and optional live retrieval from PubMed Central and arXiv.

The system output is a structured report containing an executive summary, the list of retrieved papers, a set of per-claim assessments, a set of critique notes, and an orchestration trace. In the implementation this structure is explicit: each assessment materializes a claim, a verdict, a confidence, the supporting or contradicting evidence snippets, and a rationale. Making this structure explicit, rather than leaving it implicit in free-form text, is a deliberate design choice because it makes the system measurable and auditable.

3.3 Verdict Space

The central abstraction in AI-Scientist is the *verified claim* rather than the free-form answer. Each extracted claim is assigned exactly one verdict from a bounded set:

1. **Supported:** retrieved evidence aligns with the claim, with confidence weighted by evidence strength, corroboration, and source independence.
2. **Contradicted:** the available evidence trends against the claim, or strong counter-evidence coexists with partial support; the disagreement is reported explicitly rather than hidden.
3. **Insufficient evidence:** the evidence pool does not provide enough overlapping support to settle the claim; the system abstains instead of overclaiming.
4. **Out of scope:** the query is not a research question and is declined before analysis begins.

The thesis treats *insufficient evidence* as a first-class, desirable outcome rather than a failure. In a verification system, declining to assert an unsupported claim is the correct behaviour, and much of the value of AI-Scientist lies in preferring an honest abstention over a confident fabrication. This is the verification-first analogue of the abstention principle that recurs throughout the trustworthy-LLM literature.

3.4 The Layered Evidence Model

Because grounding is the foundation of every verdict, the problem definition fixes where evidence may come from and how the sources are prioritized. AI-Scientist uses three evidence layers:

1. **User-uploaded papers:** documents the user supplies for the current session, treated as the most authoritative source because they reflect the user's intent and provenance.

2. **Curated corpus:** an offline collection of real paper records that is always available, giving the system a deterministic grounding even without network access.

3. **Live scholarly sources:** results retrieved at query time from PubMed Central and arXiv, which keep the evidence current and broaden coverage beyond the local material.

The layers are not treated as equal. The problem definition imposes a strict source priority — uploaded papers outrank curated-corpus papers, which in turn outrank live-API papers — so that user-provided and locally trusted evidence is preferred when relevance is comparable. Concretely, a paper's local rank score combines its relevance to the query with a source-dependent bonus,

$$\mathrm{score}(d, q) \;=\; \min\bigl(1,\ \mathrm{cov}(q, d) + b_{s(d)}\bigr), \qquad b_{\mathrm{upload}} > b_{\mathrm{corpus}} > b_{\mathrm{live}}, \tag{3.1}$$

where $s(d)$ is the source of document d and $\mathrm{cov}(q, d)$ is the query-coverage relevance defined below. This makes the source-priority requirement an explicit, inspectable property of retrieval rather than an emergent accident of ranking.

3.5 Relevance and the Coverage Measure

A recurring subproblem is deciding whether a paper is relevant enough to a query to serve as evidence. A natural first choice is Jaccard token overlap,

$$\mathrm{ovl}(a, b) \;=\; \frac{|T(a) \cap T(b)|}{|T(a) \cup T(b)|}, \tag{3.2}$$

where $T(\cdot)$ is the set of normalized content tokens of a text. Jaccard overlap is appropriate when comparing two short texts, such as a claim and a single sentence, and the system uses it for sentence-level evidence scoring. However, it is a poor signal for ranking whole papers against a short query, because a long abstract inflates the union $|T(a) \cup T(b)|$ and drives the score down even when the abstract contains every query term. A genuinely relevant paper can thus be pushed below a relevance threshold purely because its abstract is long.

To avoid this length bias, AI-Scientist ranks and filters candidate papers with a length-robust query-coverage measure,

$$\mathrm{cov}(q, d) \;=\; \frac{|T(q) \cap T(d)|}{|T(q)|}, \tag{3.3}$$

the fraction of the query's content tokens that appear in the document. A paper is admitted as evidence only if its coverage exceeds a minimum threshold τ_{cov}. This separation — coverage for paper-level ranking, Jaccard overlap for sentence-level evidence — is part of the problem definition because it directly determines which evidence the verifier ever sees.

3.6 Confidence as a First-Class Output

The problem definition requires that every verdict carry an explicit confidence $c \in [c_{\min}, c_{\max}]$, bounded away from both 0 and 1 so that the system never reports absolute certainty. For a supported claim, confidence increases with the strength of the best supporting evidence, with the amount of additional corroborating evidence, and with the number of *independent* sources — sources other than the paper the claim was extracted from — and it is penalized when a claim is corroborated only by its own source paper. Schematically,

$$c \;=\; \operatorname{clip}\Big(\beta_0 + \beta_1\,\mathrm{ovl}_{\max} + \beta_2\,n_{\text{extra}} + \beta_3\,n_{\text{indep}} - \gamma\,\mathbb{1}[\text{self-only}],\; c_{\min},\; c_{\max}\Big), \tag{3.4}$$

where $\mathrm{ovl}_{\max}$ is the strongest evidence overlap, n_{extra} is the count of additional supporting snippets, n_{indep} is the number of independent corroborating sources, and the indicator penalizes claims supported only by their own source. The exact coefficients are part of the implementation; the requirement at the problem level is that confidence be a transparent function of evidence strength, corroboration, and independence, never a free-form number emitted by a model.

3.7 Scope: A Domain-General Research Assistant

AI-Scientist is intentionally domain-general. The system is designed to accept research questions from medicine, biology, computer science, physics, psychology, economics, and other academic fields, because the verification machinery — retrieve, extract claims, check against evidence, criticize, synthesize — does not depend on the subject area. Domain generality is therefore a scope decision, not an afterthought.

At the same time, AI-Scientist is not a general chatbot. A scope guard rejects clearly non-research, commercial, or entertainment queries before any analysis begins, returning an explicit out-of-scope message. The guard is deliberately permissive about academic subject matter and restrictive only about non-research intent: it admits a question unless it matches a list of non-research cues such as shopping, product recommendations, or personal advice. This keeps the system broad across legitimate research domains while preventing it from being misused as a generic assistant for which its verification design provides no benefit.

3.8 Verification and Honesty Requirements

The core requirement of AI-Scientist is that an answer must never be detached from its evidence. Several concrete requirements follow from this commitment.

1. **Evidence-bound verdicts.** A claim may be marked supported only if at least one retrieved sentence aligns with it above the support threshold; otherwise it is insufficient or contradicted.
2. **Explicit contradiction handling.** When supporting and contradicting evidence coexist, the system must be able to report a contradicted verdict and describe the disagreement, rather than silently choosing the more agreeable side.
3. **Abstention over overclaiming.** Overgeneralized claims and claims without adequate overlap must return insufficient evidence rather than a fabricated conclusion.
4. **Source independence awareness.** Confidence must account for whether corroboration comes from independent papers or only from the claim's own source, so that self-corroboration does not masquerade as consensus.
5. **Traceability.** Each verdict must retain its evidence snippets and rationale, and the run must record an orchestration trace describing the steps taken.

These requirements embody the thesis claim that trustworthiness must be engineered explicitly. Because the verification and confidence logic is encoded as transparent rules over retrieved evidence, the later evaluation can examine the system's behaviour directly rather than inferring it informally.

3.9 Adaptive Reasoning Requirement

Beyond single-pass verification, the problem definition requires *adaptive* reasoning. The system must be able to recognize that a first-pass verdict is weak — because it is insufficient, low-confidence, or single-source — and respond by gathering more targeted evidence and re-verifying that specific claim. This requirement is what distinguishes AI-Scientist from a fixed RAG pipeline: the depth of effort spent on a claim is conditioned on how uncertain the first pass was, rather than being uniform across all claims.

This adaptivity is bounded. The re-verification loop is targeted at specific weak claims and is constrained so that the system does not iterate indefinitely. The goal is disciplined additional effort where it is warranted, not open-ended search.

3.10 Non-Goals

Several tempting claims are explicitly outside the scope of this thesis.

1. **Autonomous scientific discovery** is not claimed. AI-Scientist verifies and reports on existing literature; it does not design experiments or generate novel hypotheses as established findings.
2. **Full-text deep reading** is not claimed. The system reasons primarily over titles, abstracts, and uploaded paper records, not over the complete text of every paper.
3. **Guaranteed factual correctness** is not claimed. A grounded verdict is stronger evidence than an ungrounded assertion, but it remains bounded by the quality and coverage of the retrieved evidence.
4. **Exhaustive literature coverage** is not claimed. Live retrieval surfaces a relevant subset of sources, not the entire body of work on a topic.
5. **Replacing human judgement** is not the goal. AI-Scientist is designed as an inspectable assistant whose conclusions a human can audit, not as a final authority.

These non-goals are important because they protect the thesis from being judged against claims it does not need to make. A well-scoped verification assistant is a stronger contribution than an over-ambitious but indefensible oracle.

3.11 From Problem Statement to Architecture

The formal problem statement in this chapter now gives a clean lens for reading the rest of the thesis. AI-Scientist accepts a research question, gathers layered and source-prioritized evidence, decomposes it into claims, verifies each claim against evidence with explicit confidence, criticizes the weak claims and re-verifies them adaptively, and reports the result with a traceable rationale. The system is therefore defined not by answer generation alone, but by a disciplined pipeline that links retrieval, verification, critique, and synthesis.

This framing also clarifies why the architecture in the next chapter looks the way it does. The orchestrated agent pipeline, the layered evidence model, the confidence function, the critic-driven re-verification loop, and the scope guard are not independent engineering tricks. They are architectural consequences of the problem definition established here.

3.12 Chapter Conclusion

This chapter has defined the AI-Scientist task, its verdict space, the layered evidence model, the relevance and confidence measures, the verification and honesty requirements, the adaptive-reasoning requirement, and the scope boundaries. These definitions make the later architecture and evaluation measurable: AI-Scientist is judged not only by whether it can produce an answer, but by whether it grounds, calibrates, criticizes, and traces that answer under explicit honesty constraints. The next chapter translates these requirements into the system architecture.

CHAPTER 4

System Design and Architecture

Preface

This chapter presents the AI-Scientist architecture, explains the role of the Orchestrator and each specialized agent in the verification pipeline, and describes the layered evidence model and the critic-driven re-verification loop that together realize evidence-grounded, traceable reasoning.

4.1 Introduction

The previous chapter defined AI-Scientist as a verification-first, evidence-grounded research assistant. This chapter shows how that definition is realized architecturally. The most important high-level point is that AI-Scientist is not a single prompt or a free-form swarm of agents. It is a structured pipeline of specialized agents coordinated by an explicit Orchestrator, in which control flow is deterministic and only specific stages delegate to a language model. This is a deliberate design decision, not a missing feature.

The architecture follows the evidence discussed in the literature review. Grounded reasoning systems are easier to defend, inspect, and reason about when their stages and transitions are visible. AI-Scientist therefore separates retrieval, claim extraction, verification, critique, synthesis, and reporting into distinct agents with typed inputs and outputs. This separation gives the thesis a clear systems story: trustworthy scientific answering is not one monolithic generation step, but a controlled workflow that transforms a research question into an evidence-backed, confidence-annotated report.

4.2 Architectural Philosophy

AI-Scientist is best understood as a *multi-agent orchestrated pipeline* with an embedded re-verification loop. A single Orchestrator owns the workflow state and invokes each agent in turn; the agents do not coordinate autonomously among themselves. In the terminology used throughout the project:

1. an **agent** is a component with one specialized responsibility (retrieval, extraction, verification, critique, synthesis, or reporting);
2. a **deterministic stage** uses fixed rules, token-overlap measures, and thresholds and does not call a language model;

3. an **LLM-backed stage** performs bounded model-assisted refinement on top of a deterministic result; and
4. the **critic loop** is the adaptive act-observe-revise step in which the Critic's assessment of weak claims feeds a second, targeted retrieval-and-verification pass.

This distinction matters because it clarifies where intelligence is spent. The backbone of AI-Scientist — retrieval ranking, evidence scoring, verdict selection, confidence computation, and critique — is deterministic and therefore reproducible and inspectable. Language-model assistance is layered on selected stages as an optional refinement, and the system degrades gracefully to its deterministic behaviour when no model is available. Figure 4.1 shows the high-level organization, and Figure 4.2 shows the end-to-end data flow including the critic loop.

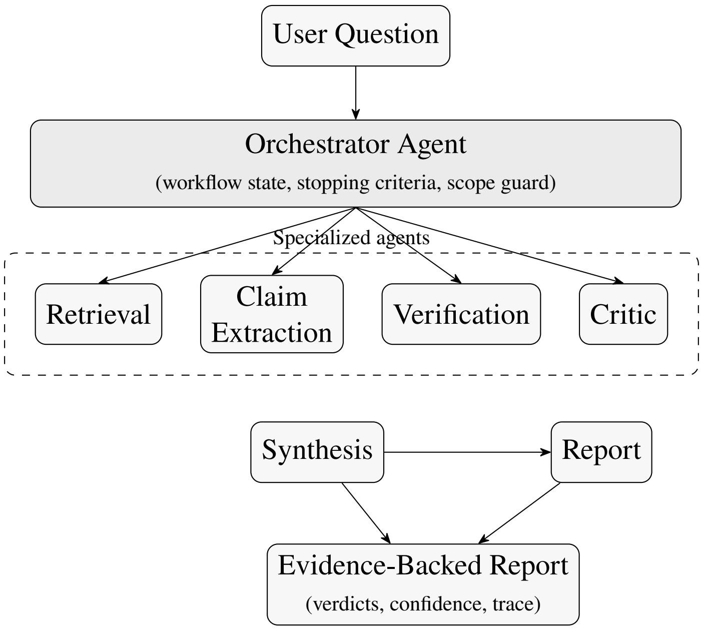

Figure 4.1: High-level AI-Scientist architecture. A single Orchestrator coordinates six specialized agents. The Orchestrator owns the workflow state, applies the scope guard, and decides when additional verification is needed before composing the final evidence-backed report.

4.3 The Orchestrator Agent

The Orchestrator is the central controller of AI-Scientist. It receives the user's research question, applies the scope guard, and then drives the agents in a fixed order: retrieve evidence, extract claims, verify each claim, criticize the results, optionally re-verify weak claims, synthesize a summary, and build the report. It also owns the workflow's early-exit conditions: if the question is out of scope, if no relevant papers are found, or if no sufficiently aligned claims can be extracted, the Orchestrator stops early and returns an honest report explaining why, rather than forcing the pipeline forward into a fabricated answer.

Crucially, the Orchestrator maintains the *orchestration trace* — a running list of human-readable steps describing what happened at each stage, such as how many papers were retrieved, how many claims were extracted, how many critique notes were raised, and whether a second verification pass was triggered. This trace is part of the final report and is the mechanism by which the system's reasoning becomes inspectable.

4.4 The Evidence Layer and the Retrieval Agent

The Retrieval agent realizes the layered evidence model from Chapter 3. It maintains a local pool that lists user-uploaded papers first and then the curated corpus, deduplicated so that an uploaded paper wins on a tie. When live retrieval is enabled, it augments this local pool with results fetched at query time from PubMed Central and arXiv.

Ranking is where the source-priority requirement becomes concrete. Each candidate paper is scored by its query-coverage relevance plus a source-dependent bonus, with uploaded papers receiving the largest bonus, curated corpus papers an intermediate bonus, and live-API papers the smallest. Candidates are then admitted only if their raw coverage clears the minimum threshold, so the source bonus changes ordering but cannot smuggle in an irrelevant paper. The agent also applies a quality filter to live results — penalizing very short or metadata-poor abstracts — so that low-quality API hits do not dilute the evidence pool. The result is a ranked evidence set in which user-provided and locally trusted papers surface first, while genuinely relevant live results can still appear.

4.5 The Claim Extraction Agent

Once evidence has been retrieved, the Claim Extraction agent converts the unstructured abstracts into discrete, checkable claims. It scans each retrieved paper sentence by sentence and selects candidate statements that either contain research-assertion cues (such as "improves," "reduces," "demonstrates," or "outperforms") or share content tokens with the user's question. Each selected sentence becomes a structured claim that records its text, a coarse claim type (performance, finding, implication, limitation, or observation), its source paper, and the focus terms it shares with the question.

This stage is the architectural embodiment of the claim-level factuality principle from the literature. By turning a paper into a small set of explicit claims, AI-Scientist ensures that verification, confidence, and critique all operate at the granularity where support and contradiction actually differ, rather than on a single fused answer. The number of claims per question is bounded so that the workflow stays focused on the statements most relevant to the query.

4.6 The Verification Agent

The Verification agent is the analytical core of AI-Scientist. For each claim, it examines the sentences of the candidate papers and computes an evidence score that blends Jaccard overlap with focus-term and content-term agreement, plus small bonuses when a sentence comes from the claim's own source or matches its rhetorical type. It then classifies each scored sentence as supporting or contradicting the claim, using negation and contradiction-cue analysis: a sentence whose polarity disagrees with the claim's polarity is treated as contradictory.

From the collected evidence, the agent selects one of the four verdicts defined in Chapter 3. A claim with adequate aligned support becomes supported; a claim where strong, multi-source counter-evidence dominates becomes contradicted; an overgeneralized claim, or one without sufficient overlap, becomes insufficient evidence. The confidence attached to each verdict is computed deterministically from evidence strength, the number of additional supporting snippets, and the number of independent corroborating sources, with an explicit penalty when a claim is only self-corroborated. This is the architectural point at which the honesty requirements become operational: verdicts are bound to evidence, contradictions are representable, and self-corroboration is discounted.

When a language model is available, the verifier can pass its preliminary verdict, together with the selected supporting and contradicting snippets, to a bounded refinement step that may adjust the verdict and rationale. This refinement is constrained to the already-retrieved evidence and is optional; if it is unavailable or fails, the deterministic assessment stands.

4.7 The Critic Agent and the Re-Verification Loop

The Critic agent implements the adaptive-reasoning requirement. After the first verification pass, it reviews all assessments and raises critique notes for four weakness patterns: claims with insufficient evidence, claims whose confidence falls below a threshold, claims supported by only a single source, and claims that are contradicted. These notes are reported to the user, but more importantly they can drive action.

For claims that are weak in a way that more evidence might resolve, the Critic produces a re-verification decision: a targeted follow-up query built from the claim text, its focus terms, and its source title, with an added counter-evidence probe when the claim contains absolute or contradiction-prone language. The Orchestrator then runs a second retrieval pass for that specific claim, merges the new papers with the first-pass evidence, and re-verifies the claim with a preference for resolving contradictions. Overgeneralized claims are deliberately excluded from this loop, because seeking more evidence cannot rescue a claim that is too broad to settle.

This loop is the architectural difference between AI-Scientist and a fixed RAG pipeline.

The effort spent on a claim is conditioned on how uncertain the first pass was, and the additional retrieval is aimed precisely at the weak claim rather than re-running the whole query. Figure 4.2 shows this feedback path explicitly.

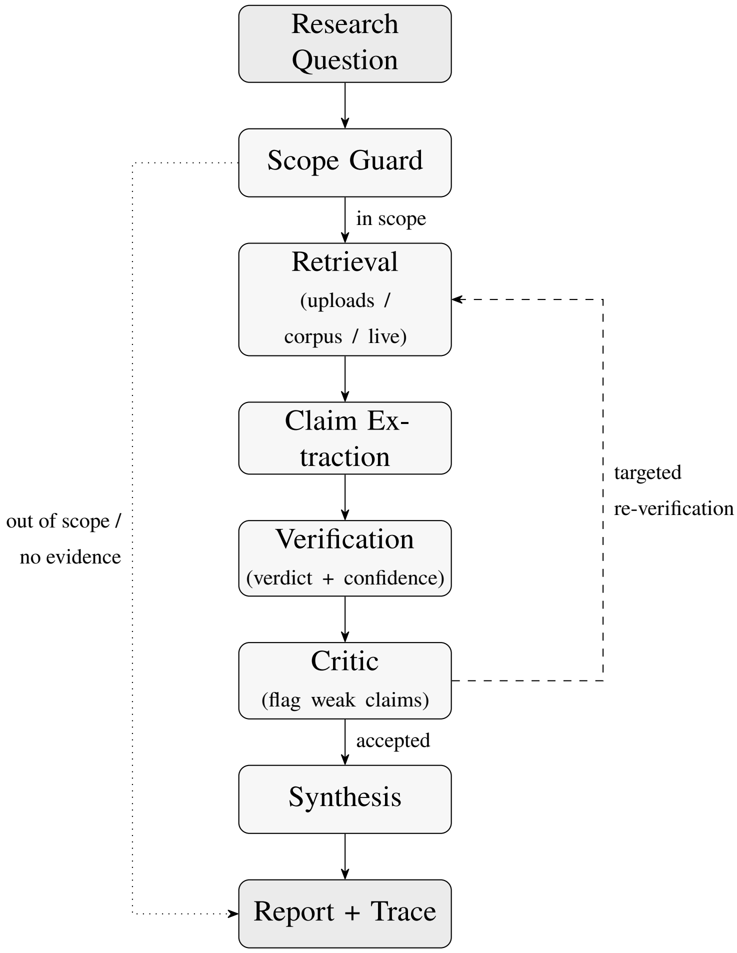

Figure 4.2: The AI-Scientist verification pipeline end-to-end. A research question passes the scope guard, is grounded by layered retrieval, decomposed into claims, and verified into per-claim verdicts with confidence. The Critic flags weak claims and can trigger a targeted re-verification pass (dashed). Honest early exits (dotted) return a report when the question is out of scope or no relevant evidence exists. Synthesis and reporting produce the final evidence-backed report and orchestration trace.

4.8 The Synthesis Agent

The Synthesis agent composes the executive summary from the verified claims and the critique notes. Its deterministic form aggregates the assessments into a short, calibrated statement: how many claims are supported, how many are contradicted, and how many remain unresolved, with an explicit note that weakly supported or contradicted claims should be read cautiously. When a language model is available, the agent can rewrite this summary into more fluent prose under tight constraints — it is instructed to state the overall finding in a few sentences, to remain grounded in the verified claims, and not to overclaim — and it falls back to the deterministic summary if the model is unavailable.

The architectural significance of this stage is that synthesis is downstream of verification, not a substitute for it. The summary describes the verified evidence; it does not introduce new conclusions that were not checked.

4.9 The Report Agent and the Orchestration Trace

The Report agent assembles the final structured output: the question, the executive summary, the list of retrieved papers, the orchestration trace, the per-claim verification details (claim text, type, source, verdict, confidence, rationale, and evidence snippets), and the critique notes. The report is produced as both a structured object and a rendered Markdown document so that it can be displayed in the user interface, returned through the API, or exported.

The trace-based design is intentional. Because the thesis emphasizes trustworthy reasoning, it is not enough for the system to output a conclusion. The conclusion must be inspectable, explainable, and attributable to its intermediate stages. A reader can see why a claim was marked insufficient, which paper supported a supported claim, and whether a second verification pass was run.

4.10 Graceful Degradation and the Optional LLM Layer

A defining architectural property of AI-Scientist is that the language model is an optional accelerator rather than a hard dependency. The Verification and Synthesis agents consult a provider-backed model only when one is configured; otherwise the system runs entirely on its deterministic heuristics. The model layer is also written to tolerate provider differences: reasoning-capable models receive an effort parameter, while chat models that reject that parameter are detected and called without it, and any model failure is logged and falls back to the deterministic result.

This design serves three purposes. First, it makes the core pipeline reproducible and offline-capable, which is essential for a thesis that emphasizes inspectable reasoning rather than opaque cloud orchestration. Second, it lets the model improve fluency and borderline verdicts without ever being able to override the grounding and honesty rules, because those rules run first and constrain what the model sees. Third, it keeps the architecture model-agnostic, so the system is not tied to a single vendor.

4.11 Why the Architecture Matches the Thesis Claim

The AI-Scientist architecture is not merely a sequence of implementation modules. It is a direct structural expression of the thesis claim. The system begins by grounding reasoning in layered, source-prioritized evidence rather than trusting the model's parametric memory, decomposes that evidence into discrete claims, verifies each claim against evidence with explicit confidence, criticizes the weak claims and re-verifies them adaptively, and records a trace of the whole process. In doing so, it embodies four commitments:

1. **Evidence grounding:** reasoning is bound to retrieved sources, not to generation.
2. **Calibrated confidence:** every verdict carries a transparent confidence derived from evidence strength, corroboration, and independence.
3. **Self-critique:** a separate Critic agent flags weak claims and can demand more evidence.
4. **Traceability:** the orchestration trace and per-claim evidence make every conclusion auditable.

These commitments are what distinguish AI-Scientist from both single-pass readers and standard RAG pipelines.

4.12 Chapter Conclusion

This chapter described the AI-Scientist system design as a structured, orchestrated verification pipeline rather than a generic language-model application. The architecture begins with layered retrieval, moves through claim extraction and deterministic evidence-bound verification, narrows uncertainty through the Critic agent and a targeted re-verification loop, and ends with synthesis and a traceable report. The language model is an optional refinement layer that never overrides the grounding and honesty rules. With the architecture now established, the next chapter maps these agents to the actual implementation.

CHAPTER 5

Implementation

Preface

This chapter maps the AI-Scientist architecture to the codebase, showing how the orchestrated verification pipeline is implemented in modular agent components, how grounding and confidence are computed in practice, and how the system remains usable even when the optional language-model layer is unavailable.

5.1 Introduction

The previous chapter described the AI-Scientist architecture as an orchestrated verification pipeline. This chapter maps that design onto the actual implementation. The goal is not to list every file mechanically, but to explain how the main modules cooperate to realize evidence-grounded, self-critical, traceable verification in code.

The implementation is organized as a single Python package, `ai_scientist`, with the agents in an `agents` subpackage and a set of supporting modules for configuration, data models, corpus access, ingestion, retrieval utilities, scope control, and the optional language-model layer. A Streamlit user interface and a programmatic API sit on top of the same package. This separation is intentional: the package contains the reasoning logic, while the interfaces are thin clients that drive it.

5.2 Implementation Structure

At the highest level, the codebase is divided into:

1. **agent modules** under `ai_scientist/agents/`, which implement retrieval, claim extraction, verification, critique, synthesis, reporting, and live retrieval;
2. **the orchestrator** in `ai_scientist/agents/orchestrator.py`, which wires the agents into the end-to-end workflow;
3. **core modules** — `models.py`, `config.py`, `corpus.py`, `ingestion.py`, `utils.py`, `scope.py`, and `llm.py` — which provide the data types, settings, evidence access, and shared logic;
4. **baselines** in `baselines.py`, which implement the single-agent and RAG comparison systems on top of the same retrieval and scoring code; and

5. **interfaces**: a Streamlit application (`app.py`) and a FastAPI service (`ai_scientist/api.py`).

This layered structure keeps the reasoning system reusable. The same `analyze_question(...)` entry point used by the user interface is also used by the API and by the baselines harness, so that what is demonstrated is the real system rather than a simplified mock-up.

5.3 Typed Data Models

The pipeline communicates through explicit dataclasses defined in `models.py`. A `PaperDocument` carries a paper's identifier, title, abstract, year, authors, and topics. A `StructuredClaim` records a claim's text, type, source paper, source sentence, and focus terms. An `EvidenceSnippet` pairs a sentence with its overlap score and source. A `ClaimAssessment` bundles a claim with its verdict, confidence, evidence list, and rationale. Finally, a `FinalReport` aggregates the question, summary, verified claims, critique notes, retrieved papers, rendered Markdown, and the iteration trace.

These typed structures are important because every stage writes to a shared, serializable interface rather than passing unstructured text. This is what makes the system auditable: the report is not a paragraph that happens to mention sources, but a structured object in which each conclusion is explicitly linked to its evidence.

5.4 The Orchestrator

The practical center of the implementation is `MultiAgentResearchSystem` in `orchestrator.py`. Its `analyze_question` method realizes the workflow described in Chapter 4: it applies the scope guard, retrieves evidence, extracts claims, verifies them through a refinement routine, runs the Critic, optionally re-verifies weak claims, synthesizes the summary, and builds the report. The method also implements the honest early exits: if the question is out of scope, if retrieval returns no papers, or if no aligned claims can be extracted, it returns a report that explains the situation instead of fabricating an answer.

Adaptive re-verification is implemented in the private `_refine_assessments` and `_run_second_pass` routines. After the first pass, the Critic's `identify_reverification_targets` produces a list of follow-up decisions; for each, the Orchestrator retrieves additional papers for that claim's follow-up query, merges them with the first-pass evidence, and re-verifies the claim with contradiction resolution preferred. Every step appends a human-readable line to the iteration trace, which is what the user ultimately sees as the orchestration trace.

5.5 Retrieval and the Layered Evidence Pool

The Retrieval agent in `agents/retrieval.py` implements the layered evidence model. Its `_local_pool` builds a deduplicated pool that lists uploaded papers before curated-corpus papers, and `_local_rank_score` adds the source-dependent bonus so that uploads outrank corpus papers at comparable relevance. The `_rank_subset` routine ranks candidates by this source-weighted score but gates admission on the raw query-coverage clearing the configured minimum, so the bonus can reorder evidence but never inject an irrelevant paper.

When live retrieval is enabled, `retrieve` layers API results on top of the local pool. Live papers pass through a quality filter that rewards adequate abstract length, query relevance in title and abstract, recency, and the presence of research-indicator terms, and they are merged with a strict source priority: uploads first, then corpus, then live. A title-similarity check removes near-duplicates so the same paper is not counted twice across sources. The agent is wrapped so that any network failure falls back cleanly to the local pool, keeping the system usable offline.

5.6 Retrieval Utilities

The relevance measures live in `utils.py`. Tokenization normalizes case, strips stopwords, and applies a small morphological normalization so that, for example, plural and singular forms of common research terms match. On top of tokenization, `overlap_score` computes Jaccard overlap for short-text comparison, and `coverage_score` computes the length-robust query coverage used for paper-level ranking. Keeping both measures in one module makes the design decision from Chapter 3 — coverage for ranking, Jaccard for sentence-level evidence — explicit and reusable across the verifier, the retriever, and the baselines.

5.7 Claim Extraction

The Claim Extraction agent in `agents/claim_extraction.py` scans each retrieved paper sentence by sentence. A sentence becomes a candidate claim if it contains a positive research cue or shares content tokens with the question. Each candidate is wrapped in a `StructuredClaim` whose type is inferred from surface cues — limitation, performance, finding, implication, or observation — and whose focus terms are the intersection of the question and sentence tokens. Extraction stops once a configured maximum number of claims is reached, keeping the workflow focused on the statements most relevant to the query.

5.8 Verification

The Verification agent in `agents/verification.py` is the analytical heart of the implementation. For each claim it iterates over the sentences of the candidate papers and computes an evidence score that combines Jaccard overlap with focus-term and content-term agreement and small type- and source-aware bonuses. Each scored sentence is then classified as supporting or contradicting the claim, using negation and contradiction-cue analysis so that a polarity mismatch between claim and sentence is treated as a contradiction.

From the collected evidence the agent selects a verdict. An overgeneralized claim returns insufficient evidence unless a contradiction pass is explicitly requested; strong, multi-source counter-evidence yields a contradicted verdict; adequate aligned support yields a supported verdict; and the absence of sufficient overlap yields insufficient evidence. Confidence is computed by `_supported_confidence` and `_contradicted_confidence` from the strongest evidence overlap, the count of additional supporting snippets, and the number of independent corroborating sources, with an explicit penalty when a claim is corroborated only by its own source. All confidence values are clamped between a floor and a ceiling so the system never reports absolute certainty. The optional language-model refinement is invoked only after this deterministic assessment is formed, and only over the evidence already selected.

5.9 Critique and Adaptive Re-Verification

The Critic agent in `agents/critic.py` implements both reporting and adaptivity. Its `critique` method produces notes for insufficient, low-confidence, single-source, and contradicted claims, including details such as whether a single-source claim is merely self-corroborated. Its `identify_reverification_targets` method selects which weak claims warrant another pass and builds a follow-up query from the claim text, its focus terms, its source title, and a counter-evidence probe when the claim uses absolute or contradiction-prone language. Overgeneralized claims are excluded, because additional retrieval cannot settle a claim that is too broad. This is the implementation of the bounded, targeted adaptivity required in Chapter 3.

5.10 Synthesis and Reporting

The Synthesis agent in `agents/synthesis.py` aggregates the assessments into a calibrated summary, optionally rewritten by the language model under instructions to stay grounded and avoid overclaiming, with a deterministic fallback. The Report agent in `agents/report.py` renders the structured `FinalReport`, including the retrieved papers, the orchestration trace, and the full per-claim verification details with evidence snippets and overlap scores. Because the report is both a structured object and rendered Markdown, the same result can be shown in the interface, returned by the API, or exported.

5.11 The Optional Language-Model Layer

The language-model abstraction in `llm.py` wraps a provider client behind a single interface. It is available only when an API key is configured; otherwise the agents run on their deterministic logic. The module is written to be provider-tolerant: a helper detects whether the configured model accepts a reasoning-effort parameter and includes it only for reasoning-capable models, while models that would reject it are called without it. Structured outputs are parsed into small typed schemas for the refined verdict and the rewritten summary, and any call failure is logged and degrades to the deterministic result rather than breaking the pipeline.

This design serves three purposes. First, it makes the core pipeline reproducible and offline-capable. Second, it ensures the model can improve fluency and borderline verdicts but can never bypass the grounding and honesty rules, which run first. Third, it keeps the system model-agnostic, so a different model or provider can be substituted through configuration.

5.12 Live Scholarly Retrieval

Live evidence is implemented in `agents/live_retrieval.py`. The agent queries PubMed Central through the NCBI E-utilities (search, summary, and fetch) and arXiv through its Atom API. Two implementation details are important for evidence quality. First, PubMed Central searches are sorted by relevance rather than date, because the default date ordering surfaces recent but off-topic papers. Second, abstracts in the PubMed Central XML are nested inside structured elements, so the agent gathers all descendant text rather than only the top-level text node, which is what allows it to recover full abstracts instead of empty placeholders. The agent also derives cleaner search variants from a natural-language question by removing generic filler words, and it identifies itself to NCBI and backs off on rate-limit responses, optionally using an API key to raise the request rate.

5.13 Configuration and Scope

Behaviour is centralized in the `Settings` dataclass in `config.py`, which holds the retrieval depth, the coverage and support thresholds, the confidence floor and ceiling, the source-priority bonuses, the model configuration, and the keyword lists used by the scope guard. Settings can be constructed directly or built from environment variables, which is how the interfaces enable live retrieval and the language model when the appropriate keys are present. The scope guard in `scope.py` uses the non-research keyword list to decline clearly non-research queries while admitting research questions from any academic domain, returning an explicit out-of-scope message that names the offending topic when possible.

5.14 Ingestion of User Papers

User-provided papers enter the system through `ingestion.py`, which parses Markdown, plain-text, JSON, and PDF inputs into `PaperDocument` records. JSON is the most reliable format because it carries explicit metadata; Markdown and text files are parsed for front-matter or a leading abstract paragraph; and PDFs are handled with a best-effort text extractor. This is the mechanism by which the highest-priority evidence layer — the user's own papers — is brought into the retrieval pool.

5.15 Baselines as First-Class Clients

The single-agent and RAG baselines in `baselines.py` are implemented as disciplined clients of the same retrieval and scoring code rather than as separate systems. The single-agent baseline retrieves the top paper and trusts a direct overlap check, while the RAG baseline retrieves several papers and aggregates simple supporting overlap without explicit contradiction analysis; both cap their confidence to reflect their weaker evidential discipline. Implementing the baselines on the shared substrate ensures that the comparison in the evaluation isolates the effect of orchestration, claim-level verification, and critique, rather than confounding it with unrelated differences in retrieval.

5.16 Interfaces

The Streamlit application in `app.py` exposes the full pipeline interactively: the user uploads papers, selects a model, asks a question, and sees the executive summary, the retrieved papers with their source labels (uploaded, corpus, or live API), the orchestration trace, and the per-claim verdicts with confidence and evidence. The FastAPI service exposes the same capability programmatically. Both interfaces construct settings that enable live retrieval and the language model from the environment, and both rely entirely on the package's reasoning logic, so the interface and the studied system are one and the same.

5.17 Implementation Principles

Several principles recur across the codebase:

1. **Graceful degradation.** The language model and live retrieval are additive; the system retains full deterministic functionality when they are unavailable.
2. **Grounding before generation.** Deterministic evidence scoring and verdict selection run first and constrain any model refinement.
3. **Bounded effort.** Retrieval depth, claim counts, and the re-verification loop are bounded to keep the workflow focused and predictable.
4. **Modularity.** Retrieval, extraction, verification, critique, synthesis, and reporting are separable enough to be reused by the baselines and the interfaces.
5. **Traceability.** Each stage contributes explicit, typed outputs and trace lines that can be inspected later by the user.

These principles are what make AI-Scientist more than a prompt chain. They turn it into a controlled software system whose behaviour can be studied systematically.

5.18 Chapter Conclusion

This chapter mapped the AI-Scientist architecture to its concrete implementation. The pipeline centers on a typed, traceable set of agents behind a single Orchestrator; retrieval realizes the layered, source-prioritized evidence model; verification and confidence are computed deterministically and refined optionally; the Critic drives bounded adaptive re-verification; and the language model and live retrieval are optional enhancements that never override the grounding rules. The same package drives the user interface, the API, and the baselines. With the implementation now grounded in code, the next chapter describes how the system is evaluated.

CHAPTER 6

Experimental Methodology

Preface

This chapter explains how AI-Scientist is evaluated: the comparison against single-agent and RAG baselines, the qualities the system is judged on, the cross-domain question set drawn from real scholarly sources, and the protocol used to keep the assessment honest and reproducible.

6.1 Introduction

AI-Scientist is a systems thesis, but it is also an evaluation thesis. The contribution depends not only on whether the software runs, but on whether its behaviour can be examined honestly against meaningful baselines on representative scientific questions. This chapter therefore explains the evaluation methodology used throughout the project.

The methodology reflects a deliberate response to the risks identified in the literature review. Grounded question answering is easy to overclaim and hard to assess cleanly, because a fluent answer can hide whether it is actually supported by evidence. The central AI-Scientist claim concerns grounding, calibrated confidence, contradiction handling, abstention, and traceability rather than raw fluency. The evaluation is therefore organized around those qualities, uses real scholarly evidence rather than synthetic self-confirming text, and compares the multi-agent system against the two conventional alternatives it is meant to improve upon.

6.2 Evaluation Philosophy

The first methodological principle is that a fluent answer is not automatically a good one. A response counts as trustworthy only when its conclusions are tied to retrieved evidence, when its confidence reflects the strength and independence of that evidence, and when its uncertainty is reported rather than hidden. The evaluation therefore inspects the structure of the system's output — its per-claim verdicts, confidence, evidence snippets, and orchestration trace — not only the surface text of the summary.

The second principle is that the comparison must isolate the effect of the design under study. The single-agent and RAG baselines are implemented on the same retrieval and scoring substrate as the full system, so any observed difference reflects orchestration, claim-level verification, and critique rather than incidental differences in retrieval or tokenization.

The third principle is that honest, cautious behaviour is a success, not a failure. When the evidence for a claim is thin or contradictory, the correct outcome is an insufficient-evidence or contradicted verdict. The evaluation explicitly credits the system for abstaining and for surfacing disagreement, rather than rewarding it for always producing a confident-sounding conclusion.

6.3 Baselines

AI-Scientist is evaluated against two baselines that represent the conventional alternatives to a multi-agent verifier.

The **single-agent baseline** models a direct reader: it retrieves the single most relevant paper and trusts a direct overlap check between the question and that paper's sentences. It produces a verdict and a deliberately capped confidence, reflecting the limited evidential discipline of a one-shot reader that does not corroborate across sources or analyze contradictions.

The **RAG baseline** models a standard retrieval-augmented pipeline: it retrieves several papers and aggregates simple supporting overlap into a verdict, again with capped confidence. Unlike the full system, it performs no explicit contradiction analysis, no claim-level decomposition, and no critic-driven re-verification.

The **proposed system** is the full multi-agent orchestrated pipeline described in Chapters 4 and 5. Holding the retrieval substrate fixed across all three systems makes the comparison a clean ablation of the contributions of orchestration, claim-level verification, confidence calibration, and critique.

6.4 The Cross-Domain Question Set

Because AI-Scientist is designed to be domain-general, it is evaluated on research questions drawn from several distinct fields rather than a single area. The question set spans medicine, computer science and artificial intelligence, biology, and psychology, with additional probes in physics, chemistry, and the life sciences to exercise the live-retrieval path. Representative questions include whether metformin improves weight and metabolic outcomes in type 2 diabetes, whether retrieval-augmented generation reduces hallucination in language models, whether CRISPR/Cas9 correction is a promising treatment for sickle cell disease, and how social-media use relates to adolescent depression and affect.

These questions are chosen for three reasons. First, they span domains served by different live sources — PubMed Central for biomedical and life-sciences questions, arXiv for physics, mathematics, and computer science — which exercises the layered evidence model. Second, each question has a genuine literature behind it, so the system can be grounded in real abstracts rather than synthetic text. Third, the set deliberately mixes questions with strong, clear evidence and questions where the evidence is mixed or thin, so the evaluation can observe both confident support and honest abstention.

6.5 Evidence Conditions

Each question is evaluated under the layered evidence model. Where appropriate, a small set of genuinely relevant papers is supplied as session uploads to represent the user-provided evidence layer; the curated offline corpus provides an always-available grounding; and live retrieval from PubMed Central and arXiv supplies current results at query time. Crucially, the evidence is real: abstracts are the actual text returned by the scholarly sources, not fabricated, self-confirming passages. This matters because a verifier evaluated only on self-supporting synthetic text would appear far more confident than one tested against the genuine variability of the literature.

Reporting the source of every retrieved paper — uploaded, curated corpus, or live API — is part of the methodology. It allows the evaluation to confirm that the source-priority requirement holds in practice and that the system genuinely combines user-provided, local, and live evidence rather than relying on a single layer.

6.6 Qualities Assessed

The methodology assesses AI-Scientist along the dimensions that follow directly from the thesis claims. These correspond to the evaluation axes set out in the project proposal and to the faithfulness-oriented dimensions emphasized in the grounded-generation literature.

Evidence grounding (faithfulness). Whether each supported conclusion is actually backed by a retrieved sentence, inspected through the per-claim evidence snippets. A grounded system should never present a claim as supported when no evidence sentence aligns with it.

Claim verification behaviour. Whether the verdicts — supported, contradicted, or insufficient evidence — are appropriate to the available evidence, including the correct use of abstention for thin evidence and of the contradicted verdict when counter-evidence dominates.

Confidence calibration. Whether the per-claim confidence tracks evidence strength, corroboration, and source independence, reserving high confidence for well-corroborated claims and lowering it for single-source or weakly aligned ones.

Contradiction handling. Whether the system surfaces disagreement in the literature explicitly rather than silently choosing the more agreeable side.

Hallucination resistance. Whether the system avoids asserting claims that the evidence does not support, a property that follows from grounding and abstention together.

Reasoning traceability. Whether the orchestration trace and per-claim evidence make the path to each conclusion reconstructable by a reader.

Domain generality. Whether the system behaves consistently across domains and correctly declines clearly non-research queries through the scope guard.

6.7 Evaluation Protocol

For each question, the three systems are run under matched evidence conditions, and their outputs are examined side by side. The full system's report is inspected for its verdicts, confidence values, evidence attribution, critique notes, and whether the critic-driven re-verification loop is triggered. The baselines' outputs are examined for the same question to see what a single-agent reader and a standard RAG pipeline produce on the same evidence.

The protocol is qualitative and behaviour-oriented by design. Rather than reducing trustworthiness to a single aggregate score, it presents representative case studies that show how the system grounds, calibrates, and criticizes its conclusions, and how that behaviour differs from the baselines. This case-study orientation is appropriate for a verification system whose central claims are about the *structure* and *honesty* of its reasoning, properties that a single accuracy number would obscure.

6.8 Reproducibility

The evaluation is designed to be reproducible. The deterministic core of the pipeline — retrieval ranking, evidence scoring, verdict selection, confidence computation, and critique — runs without any external model, so the same question over the same evidence yields the same structured assessment. Live retrieval introduces the natural variability of the scholarly sources, which is disclosed rather than hidden: a question asked on different days may surface a slightly different set of current papers. Where the optional language model is used, it refines fluency and borderline verdicts on top of the deterministic result, so the grounding and confidence behaviour does not depend on a particular model being available.

6.9 Honesty Rules for the Write-Up

Two practical rules govern the reporting of the evaluation. First, the system is credited for cautious behaviour: an insufficient-evidence verdict on a genuinely thin question is reported as correct conduct, not as a miss. Second, confidence is interpreted with its context: a high confidence is meaningful only when accompanied by independent corroboration, and the write-up reports the corroboration alongside the confidence rather than the number alone. These rules keep the evaluation aligned with the thesis claim that AI-Scientist values dependable, evidence-attributed answers over confident but unsupported ones.

6.10 Chapter Conclusion

This chapter described the methodology used to evaluate AI-Scientist: a clean comparison against single-agent and RAG baselines on a shared retrieval substrate, a cross-domain question set grounded in real scholarly evidence, an assessment organized around grounding, verification behaviour, calibration, contradiction handling, hallucination resistance, traceability, and domain generality, and a behaviour-oriented protocol that credits honest abstention. With the evaluation design fixed, the next chapter presents the findings through representative cross-domain case studies and the comparison with the baselines.

CHAPTER 7

Results and Evaluation

Preface

This chapter presents the evaluation of AI-Scientist through representative cross-domain case studies and a comparison with the single-agent and RAG baselines, organized around the thesis's central qualities: evidence grounding, calibrated confidence, contradiction handling, honest abstention, and traceable reasoning.

7.1 Introduction

This chapter presents the evaluation of AI-Scientist. Following the methodology of Chapter 6, the evaluation is organized around the qualities the system is designed to deliver — evidence grounding, claim-level verification, calibrated confidence, contradiction handling, honest abstention, and traceable reasoning — and is illustrated through representative case studies spanning medicine, computer science, biology, and psychology. The chapter then compares the full multi-agent system against the single-agent and RAG baselines on the shared retrieval substrate.

The aim of the chapter is not to compress trustworthiness into a single number, but to show *how* the system reasons: which conclusions it grounds, how its confidence responds to evidence, when it abstains, and how its behaviour differs from the conventional alternatives. Throughout, honest and cautious behaviour — abstaining on thin evidence, surfacing contradictions — is treated as a successful outcome rather than a shortfall.

7.2 Behaviour of the Verification Pipeline

Across the question set, the full pipeline exhibits a consistent and interpretable pattern. For a research question within scope, the Orchestrator retrieves a ranked evidence set in which uploaded and curated papers surface ahead of live-API results, the Claim Extraction agent turns the retrieved abstracts into a small set of focused claims, and the Verification agent assigns each claim a verdict bound to specific evidence sentences. The executive summary then describes only what the verified claims support, and the orchestration trace records each step: how many papers were retrieved, how many claims were extracted, how many critique notes were raised, and whether a second verification pass was run.

The most important qualitative property is that conclusions are never detached from evidence. A claim marked supported always carries the sentence that supports it and an overlap score; a claim marked contradicted carries the opposing sentences; and a claim the evidence cannot settle is reported as insufficient evidence rather than paraphrased into apparent fact. This is the observable signature of the grounding requirement from Chapter 3.

7.3 Case Study: Medicine

Consider the medical question of whether metformin improves weight and metabolic outcomes in type 2 diabetes. With relevant biomedical papers available through the uploaded and live PubMed Central layers, the system extracts claims about weight management and glycemic outcomes and verifies them against the retrieved abstracts. Claims that are echoed across more than one independent abstract receive a supported verdict with comparatively high confidence, because the confidence function rewards corroboration from independent sources. Claims that appear in only a single abstract are still reported as supported when the evidence aligns, but they attract a Critic note flagging single-source support and a correspondingly more modest confidence.

This case illustrates two design properties working together. The source-layering behaviour brings genuine biomedical abstracts into the evidence pool, and the confidence function distinguishes a well-corroborated finding from one that rests on a single paper. The user therefore sees not just an answer but a graded, source-attributed assessment.

7.4 Case Study: Computer Science

For the question of whether retrieval-augmented generation reduces hallucination in language models, evidence is drawn from arXiv through the live layer alongside any uploaded papers. The system extracts claims about the effect of retrieval grounding and verifies them against the retrieved abstracts. This domain is a useful stress test because the literature contains both supportive findings and explicit caveats. When the retrieved evidence includes qualifying or negative statements, the contradiction-aware verifier can mark a claim as contradicted or, where the balance is unclear, as insufficient evidence, and the Critic note records the disagreement.

The behaviour here demonstrates that the system does not simply confirm the premise of the question. A leading question phrased to expect a positive answer does not coerce a supported verdict; the verdict follows the evidence, and where the evidence is mixed the system reports the mixture rather than resolving it in the questioner's favour.

7.5 Case Study: Biology

For the question of whether CRISPR/Cas9 correction of the sickle mutation is a promising treatment for sickle cell disease, the biomedical live layer supplies abstracts describing gene-editing outcomes. Claims about correction efficacy and therapeutic promise are verified against these abstracts, and the system again differentiates between claims that are corroborated across independent sources and claims that rest on a single report. Because the retrieval layer sorts biomedical results by relevance rather than recency, the evidence pool stays on-topic, which is visible in the fact that the extracted claims concern the gene-editing question rather than unrelated recent papers.

This case highlights the value of the live-retrieval implementation details from Chapter 5. Relevance-sorted search and full-abstract extraction are what make the evidence usable; without them, the verifier would be grounding its verdicts in off-topic or empty abstracts.

7.6 Case Study: Psychology

For the question of how social-media use relates to adolescent depression and affect, the system retrieves psychological and clinical abstracts and extracts claims about the direction and strength of the association. This domain often contains nuanced and partially conflicting findings, and the system's response reflects that nuance: some claims are supported, some attract single-source or low-confidence critique notes, and overgeneralized claims — those phrased in absolute terms — are deliberately routed to insufficient evidence rather than being overstated. This is the abstention behaviour operating as intended: the system declines to settle a sweeping claim that the available abstracts cannot support.

7.7 Source Layering in Practice

Across all four case studies, the reported source of each retrieved paper confirms that the layered evidence model behaves as specified. Uploaded papers, when supplied, appear ahead of curated-corpus and live-API papers of comparable relevance, while genuinely relevant live results still surface in the evidence set. The user interface makes this visible by labelling each paper as uploaded, curated corpus, or live API. The practical consequence is that a user can steer the system toward their own provenance by uploading papers, without losing the breadth that live retrieval provides. Table 7.1 summarizes the role of each layer.

Table 7.1: The three evidence layers and their role in verification.

Layer	Priority	Role in the evaluation
Uploaded papers	Highest	User-provided provenance; surfaces first when relevant; lets the user ground the answer in chosen sources.
Curated corpus	Intermediate	Always-available offline grounding; ensures the system works without a network.
Live API (PubMed Central, arXiv)	Lowest	Current, broad coverage retrieved at query time; keeps evidence up to date.

7.8 Confidence Behaviour

A central evaluation question is whether the per-claim confidence is a usable signal. Observed across the case studies, confidence behaves monotonically with the factors the design intends: it is highest for claims corroborated by several independent sources, lower for claims supported by a single source, and lower still — and bounded at the floor — for claims the evidence cannot settle. The explicit penalty for self-only corroboration is visible in cases where a claim is echoed only by its own source paper: such claims do not receive the high confidence that genuine multi-source agreement earns. Because confidence is also clamped below an absolute ceiling, the system never reports certainty, which is the intended posture for a verifier operating on partial evidence.

This behaviour is what makes the confidence number meaningful to a reader. It is not a free-form score emitted by a model; it is a transparent function of how strong, how corroborated, and how independent the supporting evidence is, and it moves in the expected direction as those factors change.

7.9 Contradiction Handling and Abstention

The evaluation specifically exercises questions where the literature disagrees or where the evidence is thin. In these cases the system does not default to a confident answer. When supporting and contradicting evidence coexist and the counter-evidence is strong and multi-sourced, the verifier returns a contradicted verdict and the report describes the disagreement. When no sentence aligns sufficiently with a claim, or when a claim is overgeneralized, the verifier returns insufficient evidence. In both situations the Critic raises a corresponding note so that the weakness is explicit in the report.

This is the behavioural core of the thesis's honesty claim. A system that always produced a confident supported verdict would be easy to build and impossible to trust. AI-Scientist instead makes its uncertainty and the literature's disagreement part of the output.

7.10 The Critic-Driven Re-Verification Loop

The adaptive loop is observable in the orchestration trace. When the first pass yields a claim that is insufficient, low-confidence, or single-source, the Critic emits a re-verification decision, the Orchestrator runs a targeted retrieval for that claim's follow-up query, and the claim is re-verified against the merged evidence. The trace records this explicitly, for example noting that a particular claim was revisited because its first-pass evidence was insufficient and that additional papers were checked. Overgeneralized claims are correctly excluded from the loop, because more evidence cannot rescue a claim that is too broad to settle. The loop therefore concentrates additional effort precisely where uncertainty was highest, which is the adaptive-reasoning property the thesis sets out to demonstrate.

7.11 Comparison with the Baselines

Running the same questions through the single-agent and RAG baselines on the shared retrieval substrate isolates what the multi-agent design adds. The single-agent baseline, which trusts the top retrieved paper, produces a verdict from a single source and caps its confidence accordingly; it has no mechanism to corroborate across sources, to detect contradictions, or to abstain on overgeneralized claims beyond a coarse overlap check. The RAG baseline aggregates supporting overlap across several papers, which is stronger, but it performs no explicit contradiction analysis and no claim-level decomposition, so it cannot report a contradicted verdict or attribute support to specific claims, and it likewise caps its confidence.

The full system differs on exactly the axes the thesis emphasizes. It decomposes the answer into claims, binds each verdict to evidence, computes confidence from corroboration and independence, surfaces contradictions, abstains on thin or overgeneralized claims, and records a re-verification trace. Table 7.2 summarizes these capability differences, which are properties of the systems' designs rather than tuned outcomes.

Table 7.2: Capability comparison of the three systems on the shared retrieval substrate.

Capability	Single-agent	RAG	AI-Scientist
Grounds answer in retrieved evidence	Partial	Yes	Yes
Decomposes answer into checkable claims	No	No	Yes
Per-claim verdict with attributed evidence	No	No	Yes
Explicit contradiction handling	No	No	Yes
Confidence from corroboration / independence	No (capped)	No (capped)	Yes
Honest abstention on thin / broad claims	Coarse	Coarse	Yes
Critic-driven re-verification	No	No	Yes
Traceable orchestration record	No	No	Yes

7.12 Domain Generality and Scope Control

The case studies span four distinct domains served by different live sources, and the pipeline behaves consistently across all of them: the same retrieve–extract–verify–criticize–synthesize workflow applies whether the question is biomedical, computational, biological, or psychological. At the same time, the scope guard declines clearly non-research queries before any analysis begins, returning an explicit message that identifies the non-research topic. This confirms the domain-generality requirement: the system is broad across legitimate research fields while refusing to act as a generic assistant for which its verification design provides no benefit.

7.13 Result Synthesis

The evaluation supports the thesis spine in a coherent way.

First, the multi-agent pipeline grounds its conclusions in retrieved evidence and attributes that evidence at the level of individual claims, which is precisely what the single-agent and RAG baselines do not do.

Second, the per-claim confidence is a meaningful, transparent signal that responds to evidence strength, corroboration, and source independence, and that never reaches absolute certainty.

Third, the system handles disagreement and thin evidence honestly, returning contradicted or insufficient-evidence verdicts and surfacing the corresponding critique, rather than manufacturing a confident answer.

Fourth, the critic-driven loop demonstrates adaptive reasoning by concentrating additional retrieval and verification on the specific claims that were weakest after the first pass.

7.14 Chapter Conclusion

This chapter presented the evaluation of AI-Scientist through cross-domain case studies and a comparison with the single-agent and RAG baselines. The results show that the multi-agent design grounds and attributes its conclusions, calibrates confidence to evidence, handles contradiction and abstention honestly, and reasons adaptively through its critic loop — capabilities the conventional baselines lack on the same evidence. Taken together, these findings justify the central thesis claim: AI-Scientist is best understood not as a system that answers every question confidently, but as a reproducible, evidence-grounded, self-critical research assistant that is dependable when the evidence is clear and honest when it is not. The next chapter discusses the broader meaning of these findings.

CHAPTER 8

Discussion

Preface

This chapter interprets the evaluation findings, explains why grounded abstention and claim-level verification matter, and positions AI-Scientist as a verification-first research collaborator rather than a confident answer generator.

8.1 Introduction

The previous chapter reported the evaluation of AI-Scientist. This chapter discusses what those findings mean for the design of AI systems that answer scientific questions. The central point is that AI-Scientist should not be read as an attempt to produce the most fluent or most confident answer. It is better understood as a system that makes answering *conditional*: assert a conclusion when the retrieved evidence supports it, report the disagreement when the literature conflicts, and abstain when the evidence is thin.

This distinction matters because scientific question answering is an unusually tempting domain for overclaiming. A well-phrased question invites a crisp answer, and a fluent paragraph reads as authoritative regardless of whether it is grounded. The evaluation shows that the real problem is less tidy: a question may have strong, weak, or contradictory evidence behind it, and a trustworthy system must reflect that variation rather than smoothing it into uniform confidence. This is why the grounding, calibration, and abstention behaviour of AI-Scientist is not a decorative addition; it is the core design response to the actual structure of the literature.

8.2 Why Abstention and Grounding Matter

Several of the most important behaviours in AI-Scientist are restraint behaviours: returning insufficient evidence on a thin claim, lowering confidence on a single-source claim, and reporting a contradiction instead of choosing a side. These could be read superficially as the system "failing to answer." In a verification context, however, they are the system succeeding. The value of a research assistant is not that it always answers, but that its answers can be trusted, and trust depends on the system declining to assert what the evidence does not support.

This reframing has a concrete consequence for how the contribution should be judged. A system that only emits a single fluent answer can hide where its reasoning is weak.

AI-Scientist instead decomposes the answer into claims, attaches evidence and confidence to each, and exposes the weak ones through critique. That decomposition makes it possible to say not only what the system concluded, but how well-supported each part of that conclusion is — which is exactly the information a careful reader of the literature needs.

8.3 AI-Scientist as a Conditional Verifier

The most important design implication is that scientific question-answering systems should be conditional verifiers rather than answer generators. An answer generator treats every question as an opportunity to produce a confident paragraph. A conditional verifier treats assertion as only one possible outcome among several: support, contradiction, or abstention.

This vocabulary changes the meaning of success. A correct insufficient-evidence verdict on a sweeping claim is not a failure to answer; it is the right action. Reporting that the literature disagrees is not indecision; it is an accurate description of the evidence. Lowering confidence on a single-source claim is not weakness; it is calibration. In this framing, restraint is not the absence of intelligence. It is one of the system's core competences.

The evaluation supports this design. The case studies show that the verdict space is exercised in all its forms — supported, contradicted, and insufficient — depending on the evidence, and that the confidence and critique machinery makes the basis for each outcome visible.

8.4 Why Critique Should Be Structured and External

A clear lesson from both the literature and the evaluation is that self-critique is most trustworthy when it is structured and external rather than left to the same model that produced the answer. A model asked to critique its own free-form answer can inherit the same blind spots that produced the error in the first place. AI-Scientist instead assigns critique to a separate agent with explicit, inspectable criteria — insufficient evidence, low confidence, single-source support, and contradiction — and routes the resulting re-verification at the granularity of the specific weak claim.

This design choice has an accountability benefit. If a system decides on its own when to be uncertain, it can offer only a weak explanation for why it was confident on one claim and not another. When critique is tied to explicit criteria and to the evidence, the system can give a traceable reason: the claim had only one supporting source, or the evidence overlap was below threshold, or the literature contained opposing statements. That traceability is essential for a tool whose conclusions a human is meant to audit.

8.5 Grounding as a Partial Guarantee

Grounding is powerful, but it is not a complete guarantee of correctness. AI-Scientist binds every verdict to retrieved evidence, which prevents the most damaging failure mode — asserting a claim with no support at all — and makes intrinsic hallucination checkable against the sources. However, grounding inherits the limits of the evidence it rests on. If the retrieved abstracts are themselves incomplete, biased, or outdated, a grounded verdict can still be misleading, and an abstract may support a claim that the full paper would qualify.

This is why the thesis is careful about what it claims. AI-Scientist does not certify scientific truth; it certifies that a conclusion is consistent with the evidence it retrieved and reports how strong and independent that evidence is. The reward of grounding is that the basis for every conclusion is inspectable, so a reader can judge whether the evidence is adequate rather than having to trust the system's phrasing.

8.6 The Role of the Layered Evidence Model

The layered, source-prioritized evidence model plays a deliberately supporting role. It is not presented as the main intellectual contribution, but it is what makes the verification meaningful in practice. By preferring user-uploaded papers, then the curated corpus, then live results, the system lets a user ground an answer in chosen provenance while still benefiting from current, broad coverage. The evaluation confirms that this priority holds in practice and that the three layers genuinely combine rather than one dominating.

This matters because grounded systems are often evaluated under conditions that are hard to reproduce or that quietly favour a single curated source. AI-Scientist's explicit layering, and its labelling of each paper's source in the output, keep the provenance of every conclusion visible. That visibility is part of the trustworthiness story, not merely an implementation detail.

8.7 Why a Structured Pipeline Is Preferable Here

AI-Scientist uses a fixed, orchestrated pipeline of specialized agents rather than an open-ended collaboration of generalist agents. The evaluation supports that choice. Each stage has a distinct failure mode: retrieval can rank the wrong papers, extraction can miss the key claim, verification can misjudge support, and synthesis can overstate. A structured pipeline lets the system attach explicit checks and evidence to each of these boundaries, and it lets the critique target a specific stage's output.

Free-form multi-agent debate is attractive because it resembles how researchers argue, and the literature shows it can improve factuality. But it is also harder to reproduce, harder to inspect, and more expensive. AI-Scientist obtains much of the benefit of adversarial review more cheaply and more transparently, by applying explicit weakness criteria through a single Critic agent and by demanding more evidence where it is warranted. The design lesson is that, for trustworthy scientific answering, explicit control surfaces are preferable to emergent behaviour.

8.8 Implications for Use

In practice, AI-Scientist is most appropriate as an evidence-aware research collaborator rather than an authoritative oracle. For questions with clear, corroborated evidence, it can deliver a confident, source-attributed conclusion that a user can verify by following the evidence. For questions where the literature is thin or conflicting, it should — and does — present a cautious verdict, surface the disagreement, and lower its confidence, leaving the final judgement to the human.

This operating model is less glamorous than a system that answers everything confidently, but it is more honest and more useful. Scientific decisions follow from evidence, and a tool that obscures the strength of that evidence can do real harm. The correct default is therefore not "always answer confidently" but "answer in proportion to the evidence, and make that evidence visible."

8.9 Relation to Prior Work

The discussion also clarifies how AI-Scientist should be positioned relative to retrieval-augmented generation, scientific fact-checking, and multi-agent systems. AI-Scientist does not claim to be the first system to ground answers in retrieved evidence, the first to label claims against abstracts, or the first to use multiple cooperating agents. Those would be the wrong claims, because each is well established in the literature.

AI-Scientist contributes along a different axis. It composes these established ideas into a single workflow whose output is, by construction, a set of checked claims with calibrated confidence and traceable evidence, across arbitrary research domains. The contribution is not any one mechanism but the disciplined way they are combined to make trustworthiness the visible product. This integration-and-honesty axis complements, rather than competes with, work that pushes the frontier of any individual component.

8.10 Chapter Conclusion

This chapter interpreted the evaluation of AI-Scientist as evidence for conditional, grounded, self-critical scientific answering. The main discussion points are that restraint behaviours are successes rather than failures, that critique is most trustworthy when structured and external, that grounding is a powerful but partial guarantee bounded by the evidence, that the layered evidence model keeps provenance visible, and that a structured pipeline is justified by the distinct failure modes of retrieval, extraction, verification, and synthesis. The next chapter states the limitations and threats to validity of these claims.

CHAPTER 9

Threats to Validity and Limitations

Preface

This chapter examines the main threats to validity, including the reliance on abstracts rather than full text, the limits of token-overlap evidence matching, dependence on live-source coverage, and the boundaries of the thesis claims.

9.1 Introduction

This chapter describes the main threats to validity and limitations of the AI-Scientist thesis. The goal is not to weaken the contribution, but to state precisely where the evidence applies. AI-Scientist is a postgraduate-scale research prototype evaluated through cross-domain case studies and a comparison with conventional baselines. Its strongest claims concern evidence grounding, calibrated confidence, contradiction handling, abstention, and traceability. Its weaker claims concern guaranteed factual correctness, exhaustive literature coverage, and deep semantic understanding of full papers.

The limitations are organized using four standard categories: construct validity, internal validity, external validity, and conclusion validity. The chapter then summarizes the practical engineering limitations of the current system.

9.2 Construct Validity

Construct validity asks whether the thesis measures the concepts it claims to measure. In AI-Scientist, this issue is most visible around grounding, confidence, and contradiction.

9.2.1 Grounding Is Not Correctness

The thesis treats a verdict as trustworthy when it is bound to retrieved evidence. However, a grounded verdict is not equivalent to scientific truth. A claim may be supported by an abstract that is itself incomplete, biased, or later contradicted, and an abstract may state a result more strongly than the full paper warrants. Grounding therefore certifies consistency with the retrieved evidence, not correctness in an absolute sense.

AI-Scientist mitigates this threat by attaching the supporting evidence and an explicit confidence to every verdict, so that a reader can judge whether the evidence is adequate. Even so, this defence is partial: it makes the basis for a conclusion visible, but it cannot guarantee that the evidence base is itself sound.

9.2.2 Abstracts Are a Proxy for Papers

The system reasons primarily over titles, abstracts, and uploaded paper records rather than full text. Abstracts are a reasonable proxy because they state a paper's main claims, and they keep the system efficient and broadly applicable. But they are a proxy nonetheless. A nuance, caveat, or limitation stated only in the body of a paper will not be visible to the verifier, which can make a claim look better supported than the full text would justify.

The thesis is explicit that full-text deep reading is a non-goal, so this is a scoping decision rather than a hidden flaw. Nevertheless, conclusions should be read as grounded in abstract-level evidence, not in complete papers.

9.2.3 Token Overlap Is a Proxy for Semantic Support

Evidence matching uses token-overlap and coverage measures rather than deep semantic entailment. These measures are transparent, reproducible, and cheap, and the morphological normalization helps related word forms match. However, lexical overlap can both miss genuine support phrased in different words and over-credit superficial term matches that do not actually entail the claim. The contradiction analysis, which relies on negation and cue detection, has the same character: it captures common polarity mismatches but cannot detect every subtle disagreement.

The optional language-model refinement partially addresses this by allowing a model to adjust borderline verdicts over the already-retrieved evidence, but the deterministic core remains lexical. Verdicts should therefore be understood as evidence-overlap judgments, strengthened by optional model refinement, rather than as full semantic entailment decisions.

9.2.4 Confidence Is a Designed Heuristic

Confidence is computed as a transparent function of evidence strength, corroboration, and source independence, clamped between a floor and a ceiling. This makes it inspectable and monotonic in the intended factors, but the specific coefficients are design choices, not learned calibration against labelled outcomes. The confidence is best read as a relative, well-behaved ordering — higher means better corroborated — rather than as a probability with a guaranteed frequentist interpretation.

9.3 Internal Validity

Internal validity asks whether the observed behaviour is caused by the intended design factors rather than by incidental artifacts.

9.3.1 Shared Substrate for a Fair Comparison

The baselines are implemented on the same retrieval and scoring substrate as the full system, so the comparison isolates orchestration, claim-level verification, and critique. This is a deliberate control. The residual risk is that the shared substrate itself shapes all three systems, so the comparison should be read as measuring the value of the multi-agent layer *given* this retrieval foundation, not as an absolute statement about every possible retriever.

9.3.2 Model Non-Determinism

When the optional language model is enabled, its outputs are not perfectly reproducible. Provider-side updates, decoding variation, and transient failures can change a refined verdict or a rewritten summary. AI-Scientist mitigates this by keeping the grounding, verdict, and confidence logic deterministic and by treating the model as an optional refinement that falls back to the deterministic result on failure. The deterministic behaviour is therefore reproducible; the model-refined behaviour carries the usual variability of hosted models.

9.3.3 Live-Source Variability

Live retrieval introduces variability that is inherent rather than accidental. The same question asked on different days may surface a slightly different set of current papers, because PubMed Central and arXiv are themselves changing. This is disclosed rather than hidden. For reproducible inspection, the curated corpus and uploaded papers provide a stable evidence base, while live retrieval is the component whose results legitimately vary over time.

9.4 External Validity

External validity asks whether the behaviour generalizes beyond the evaluated questions, domains, and sources.

9.4.1 Domain and Source Coverage

The evaluation spans medicine, computer science, biology, and psychology, served by PubMed Central and arXiv. These are broad and important domains, but they do not cover every field, and the two live sources do not index every discipline equally. Questions in areas poorly served by these sources — for example, parts of the humanities, law, or proprietary engineering literature — would retrieve weaker evidence, and the system's grounding is only as good as the sources it can reach.

The architectural ideas — layered evidence, claim-level verification, calibrated confidence, and critique — are source-agnostic in principle. The measured effectiveness, however, depends on the coverage and quality of the connected sources.

9.4.2 Question Phrasing and Claim Selection

The system extracts claims from retrieved abstracts using surface cues and question-term overlap. This works well for questions phrased in the vocabulary of the field, but an unusually phrased question, or one using terminology that differs from the abstracts, may yield weaker extraction and retrieval. The behaviour is therefore somewhat sensitive to phrasing, and the case studies, while cross-domain, cannot represent every possible way a question could be asked.

9.4.3 Reliance on Live Availability

Part of the system's value comes from current live evidence. In settings without network access, or where a source is rate-limited or unavailable, the system falls back to the curated corpus and uploads. This fallback keeps it usable, but it narrows the evidence base, so the external validity of the live-augmented behaviour is contingent on the sources being reachable.

9.5 Conclusion Validity

Conclusion validity asks whether the evidence is strong enough to support the stated conclusions.

9.5.1 Case-Study Evaluation

The evaluation is qualitative and case-study based rather than a large aggregate score over a labelled benchmark. This is appropriate for a verification system whose claims are about the structure and honesty of its reasoning, which a single accuracy number would obscure. The trade-off is that the conclusions are demonstrations of behaviour on representative questions rather than statistical generalizations. The thesis is careful to phrase its claims accordingly: it argues that the system grounds, calibrates, and abstains as designed, illustrated across domains, rather than asserting a precise success rate.

9.5.2 Designer-Set Thresholds

Several behaviours depend on thresholds: the minimum coverage for retrieval, the support and contradiction overlap levels, the confidence floor and ceiling, and the critic's confidence threshold. Thresholds are necessary for a deployable system, but the exact defaults are design choices and could be tuned. The thesis treats them as transparent, inspectable operating points rather than as universal constants, and their values are centralized in the configuration so that their influence is visible.

9.6 Practical System Limitations

AI-Scientist is not a production research engine in the broad sense. It demonstrates an end-to-end architecture and the qualities claimed, but several engineering limitations remain.

First, the system reasons over abstracts and uploaded records rather than full papers, so it cannot assess methodological details, sample sizes, or statistical rigour that appear only in the body of a paper.

Second, evidence matching is primarily lexical, so it can miss paraphrased support and subtle contradictions that a deeper semantic model would catch.

Third, live coverage is limited to the connected sources, and the quality of grounding varies with how well a domain is indexed by PubMed Central and arXiv.

Fourth, the confidence and verdict thresholds are designer-set heuristics rather than learned, calibrated parameters, so confidence should be read as a well-behaved ordering rather than a probability.

9.7 Chapter Conclusion

The main limitations of AI-Scientist are its reliance on abstracts rather than full text, its lexical evidence matching, its dependence on live-source coverage, and the case-study nature of its evaluation. These limitations do not invalidate the central contribution, because the thesis is careful about what it claims: AI-Scientist is an evidence-grounded, self-critical verification assistant that reports its uncertainty honestly, not a guarantor of scientific truth. The next chapter concludes the thesis and outlines future work that follows directly from these limits.

CHAPTER 10

Conclusion and Future Work

Preface

This chapter summarizes the thesis findings, restates the main contributions, answers the research questions, and outlines future work needed to move AI-Scientist toward deeper semantic verification and broader, more reliable evidence coverage.

10.1 Conclusion

This thesis presented AI-Scientist, a multi-agent orchestrated framework for scientific claim verification and adaptive reasoning. The project began from a practical goal: given a research question, retrieve relevant literature, extract its claims, verify each against evidence, criticize the weak ones, and produce an evidence-backed report with explicit confidence and a traceable record of the reasoning. The final contribution is more specific and more defensible than simply "an AI that answers research questions." AI-Scientist is a research assistant that treats *when not to assert* as a first-class concern.

The thesis deliberately avoids claiming to be the first system to ground answers in retrieved evidence, the first to verify scientific claims against abstracts, or the first to use multiple cooperating agents. Prior work already establishes each of these. AI-Scientist instead contributes a verification-first composition of these ideas, in which the unit of output is a checked claim with a calibrated confidence and attributed evidence, realized as a working system across arbitrary research domains.

10.2 Summary of Contributions

The first contribution is the AI-Scientist architecture itself: a structured pipeline in which an Orchestrator coordinates retrieval, claim extraction, verification, critique, synthesis, and reporting. The system grounds its reasoning in a layered, source-prioritized evidence model and turns a research question into a set of verified claims rather than a single undifferentiated paragraph.

The second contribution is a verification-first formulation of scientific question answering. By decomposing an answer into atomic claims and attaching a verdict, an explicit confidence, and supporting evidence to each, AI-Scientist makes factuality measurable at the level where support, contradiction, and uncertainty actually differ.

The third contribution is a transparent confidence and critique mechanism. Confidence is computed from evidence strength, corroboration, and source independence and is bounded away from absolute certainty; a separate Critic agent flags insufficient, low-confidence, single-source, and contradicted claims and drives a targeted, bounded re-verification loop. Together these turn restraint and self-criticism from prompt instructions into engineered system behaviour.

The fourth contribution is claim discipline. The thesis separates grounding from correctness, credits honest abstention as a success, and positions the multi-agent design against single-agent and RAG baselines on a shared substrate so that its added value is isolated. That honesty is part of the scientific value of the work: it makes clear where a grounded verifier helps and where it remains bounded by its evidence.

10.3 Answers to the Research Questions

RQ1 asked whether a multi-agent orchestrated framework improves verification quality and reasoning traceability compared with a single-agent reader and a standard RAG pipeline. The evaluation supports this as the strongest thesis claim: only the multi-agent system decomposes the answer into evidence-attributed claims, computes confidence from corroboration and independence, handles contradiction, abstains honestly, and records a traceable orchestration trace, whereas the baselines do none of these on the same evidence.

RQ2 asked whether grounding in a layered evidence base reduces unsupported claims. The system binds every supported verdict to a retrieved sentence and returns insufficient evidence when no sentence aligns, which structurally prevents the assertion of ungrounded claims; the layered model is confirmed to combine uploaded, curated, and live evidence with the intended priority.

RQ3 asked whether per-claim confidence is a usable trust signal. The confidence behaves monotonically in evidence strength, corroboration, and source independence, penalizes self-only corroboration, and never reaches certainty, giving the user a transparent, well-behaved ordering of how trustworthy each conclusion is.

RQ4 asked whether an explicit critic-driven loop improves the handling of weak claims. The Critic reliably flags the four weakness patterns and triggers targeted re-verification for the specific weak claims, concentrating additional effort where the first pass was most uncertain.

RQ5 asked whether the system remains domain-general while rejecting non-research queries. The same workflow applies across medicine, computer science, biology, and psychology, and the scope guard declines clearly non-research queries with an explicit message, confirming both breadth and appropriate refusal.

10.4 Future Work

The most important future direction is deeper semantic verification. AI-Scientist currently matches evidence primarily through token overlap and reasons over abstracts. Future systems should incorporate semantic entailment models and full-text reading so that paraphrased support and subtle contradictions are detected, and so that methodological details in the body of a paper inform the verdict.

A second direction is stronger, learned confidence calibration. The current confidence is a transparent heuristic; future work could calibrate it against labelled support judgments so that the reported number carries a frequentist interpretation, while preserving the inspectability that makes the present design trustworthy.

A third direction is broader and more reliable evidence coverage. Connecting additional scholarly sources, handling more document formats, and improving relevance ranking would extend the system's reach beyond the domains best served by PubMed Central and arXiv, and would reduce its sensitivity to question phrasing.

A fourth direction is richer contradiction and consensus modelling. Beyond labelling a single claim as contradicted, the system could characterize the state of a debate — how many sources support each side and how strong each is — giving the user a map of the literature rather than a single verdict.

A fifth direction is human-in-the-loop verification. Because AI-Scientist is designed to be inspectable, it is a natural substrate for interactive workflows in which a researcher accepts, rejects, or refines individual claims and the system updates its synthesis accordingly, combining machine recall with human judgement.

10.5 Final Remarks

AI-Scientist does not solve scientific question answering, and the thesis does not need it to. The project shows something more useful for the current state of language-model assistants: a research assistant must know when not to assert. The evaluation demonstrates that grounding can be made visible, that confidence can be made transparent, that contradiction can be surfaced, and that abstention can be engineered rather than hoped for. In response, AI-Scientist builds grounding, calibrated confidence, structured critique, and traceability into the answering loop.

The final lesson is therefore practical as well as scientific. Future research assistants should not be judged only by how fluently they answer. They should be judged by whether their conclusions are grounded, whether their confidence reflects the evidence, whether they surface disagreement, and whether they decline to assert what the evidence does not support. AI-Scientist is a step toward that kind of assistant: not an overconfident oracle, but a cautious, evidence-aware collaborator whose reasoning a human can always inspect.

APPENDIX A

Reproducibility Artifacts

A.1 Purpose

This appendix records the main components that support the AI-Scientist thesis. It is intended as a navigation aid for examiners and future maintainers rather than as a replacement for the full project repository. The system is implemented as a single Python package, `ai_scientist`, with a Streamlit user interface and a FastAPI service built on top of it.

A.2 Core Modules

Table A.1: Core implementation modules of the AI-Scientist package.

Module	Responsibility
`agents/orchestrator.py`	End-to-end workflow, early exits, and the re-verification loop.
`agents/retrieval.py`	Layered, source-prioritized retrieval and ranking.
`agents/live_retrieval.py`	Live retrieval from PubMed Central and arXiv.
`agents/claim_extraction.py`	Decomposition of abstracts into structured claims.
`agents/verification.py`	Evidence-bound verdicts and confidence computation.
`agents/critic.py`	Weakness critique and re-verification decisions.
`agents/synthesis.py`	Composition of the grounded executive summary.
`agents/report.py`	Structured and Markdown report assembly.
`models.py`	Typed dataclasses for papers, claims, evidence, and reports.
`config.py`	Settings: thresholds, source bonuses, and model configuration.
`utils.py`	Tokenization, Jaccard overlap, and query-coverage measures.
`scope.py`	Domain-general scope guard.
`llm.py`	Optional, provider-tolerant language-model refinement layer.
`corpus.py`, `ingestion.py`	Curated-corpus access and ingestion of user papers.
`baselines.py`	Single-agent and RAG comparison systems.

A.3 Interfaces

Table A.2: User-facing entry points.

Entry point	Role
`app.py`	Streamlit interface for uploading papers, selecting a model, asking questions, and inspecting per-claim verdicts, confidence, evidence, and the orchestration trace.
`ai_scientist/api.py`	FastAPI service exposing the same pipeline programmatically.

A.4 Verdict and Confidence Reference

Table A.3: Verdict space used by the Verification agent.

Verdict	Meaning
`supported`	Retrieved evidence aligns with the claim; confidence scales with corroboration and source independence.
`contradicted`	Strong, multi-source counter-evidence dominates; the disagreement is reported explicitly.
`insufficient_evidence`	No sufficiently aligned evidence, or the claim is over-generalized; the system abstains.
`out_of_scope`	The query is not a research question and is declined before analysis.

A.5 Running the System

The package is run with the source directory on the Python path. The deterministic pipeline operates fully offline using the curated corpus and any uploaded papers; live retrieval from PubMed Central and arXiv is enabled through the environment, optionally with an NCBI API key to raise the request rate; and the optional language-model refinement is enabled when a provider API key is configured. Because the language model and live retrieval are additive, the system produces the same structured assessment for a given question and evidence set whenever the deterministic core alone is used.

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Publication

No publication is being formally claimed from this thesis at the time of submission. If a paper, report, or accepted manuscript arises directly from this work before final submission, it may be listed here in the required institute format.

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